Week-2

Develop an application that receives user‟s name, contact and city and displays the same using Layout Managers and Event Listeners.

Android app in Java using Android Studio that receives the user's **name**, **contact**, and **city** via `EditText` fields and displays them in a `TextView` when a button is clicked.

Using **LinearLayout** as the layout manager and implement an **OnClickListener** for the button.

activity_main.xml

layout with 3 EditTexts, a Button, and a TextView to display the result.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:orientation="vertical">

    <EditText
        android:id="@+id/n"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:inputType="textPersonName" />

    <EditText
        android:id="@+id/ph"
```

```
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Contact"
        android:inputType="phone" />

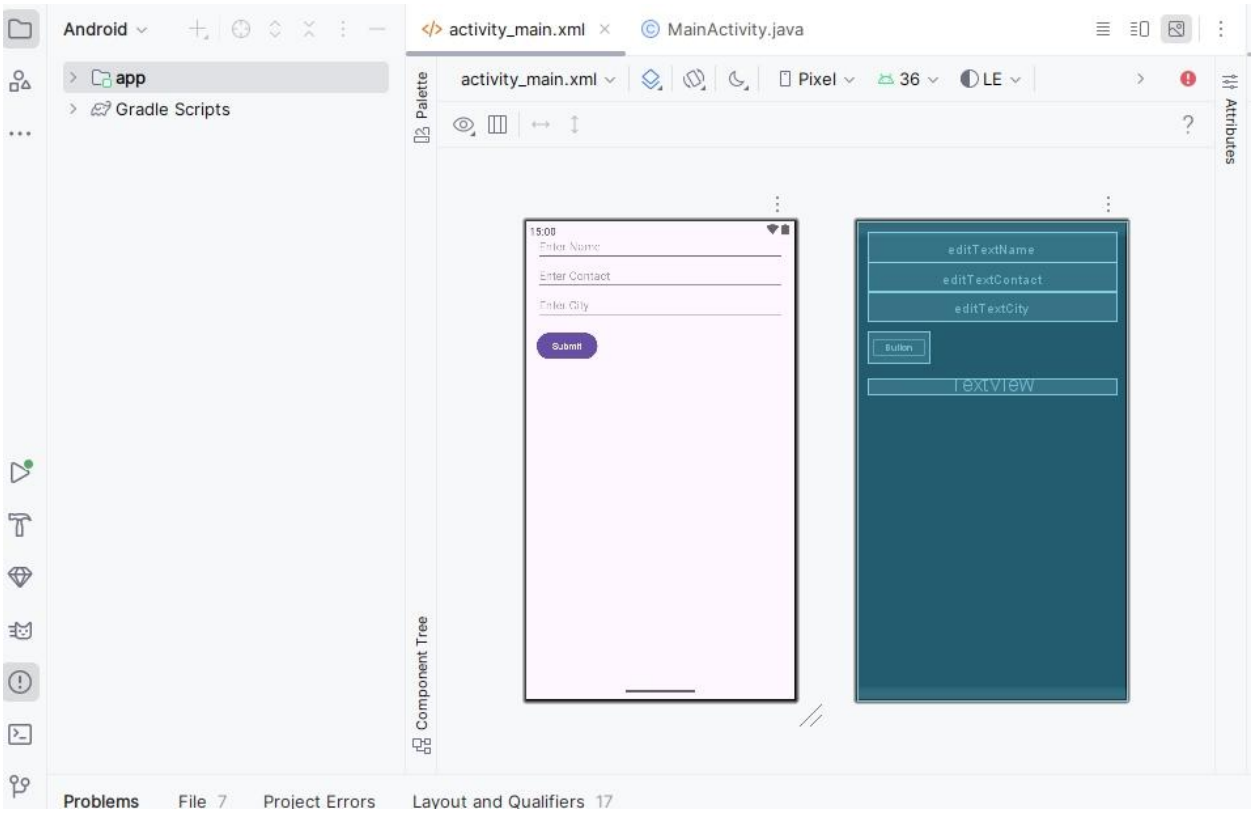
    <EditText
        android:id="@+id/ct"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter City"
        android:inputType="text" />

    <Button
        android:id="@+id/b"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"
        android:layout_marginTop="16dp" />

    <TextView
        android:id="@+id/tv"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text=""
```

```
        android:textSize="18sp"
        android:layout_marginTop="24dp" />


</LinearLayout>
```



MainActivity.java

```
package com.example.userinputapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.TextView;

public class MainActivity extends AppCompatActivity {

   EditText n, ph, ct; Button b;
   TextView tv;

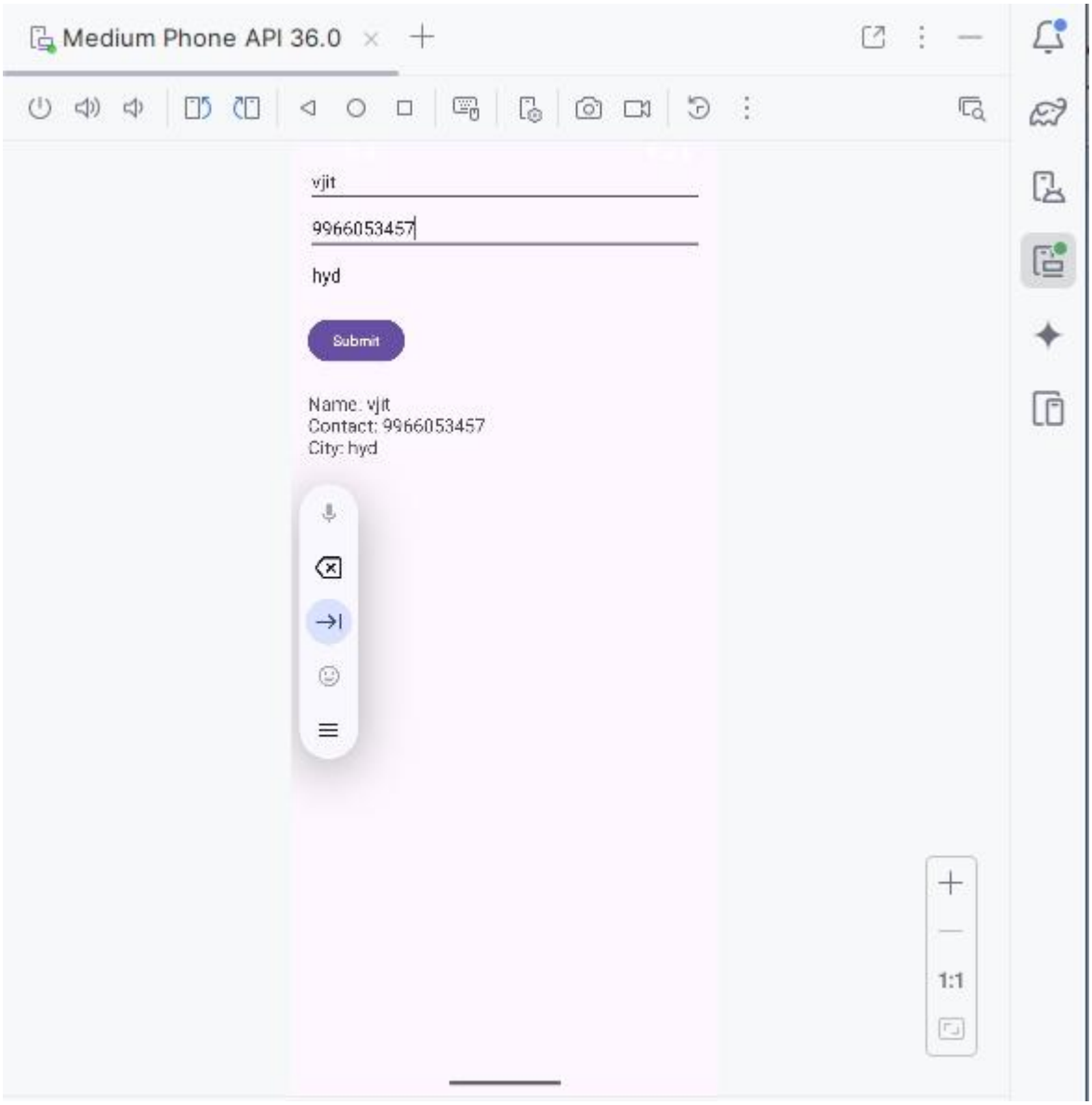
   @Override
   protected void onCreate(Bundle bd) {
super.onCreate(bd);
       setContentView(R.layout.activity_main);

        // Initialize views
       n = findViewById(R.id.n); ph = findViewById(R.id.ph); ct =
       findViewById(R.id.ct); b = findViewById(R.id.b); tv =
       findViewById(R.id.tv);
```

```
        // Set listener for button
        b.setOnClickListener(new View.OnClickListener() { @Override
            public void onClick(View v) {
                // Get input values
                String nm = n.getText().toString().trim();
                String con = ph.getText().toString().trim(); String cty =
ct.getText().toString().trim();


                // Display the values
                String msg = "Name: " + nm + "\nContact: " + con + "\nCity: " + cty; tv.setText(msg);
            }
        });
    }
}
```

Output:


Medium Phone API 36.0
vjit
9966053457
hyd
Submit
Name: vjit
Contact: 9966053457
City: hyd
1:1

Explanation:

- **Layout Manager:** `LinearLayout` arranges the views vertically.
- **Event Listener:** `b.setOnClickListener` listens for the button click.
- When the button is clicked, it fetches the input values and shows them in the `TextView`.

Develop an application that shows names as a list and on selecting a name it should show the details of the candidate on the next screen with a “Back” button.

create a simple **two-screen Android app** in Java using **Android Studio.**

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <ListView
      android:id="@+id/lv"
      android:layout_width="match_parent"
      android:layout_height="match_parent" />

</LinearLayout>
```

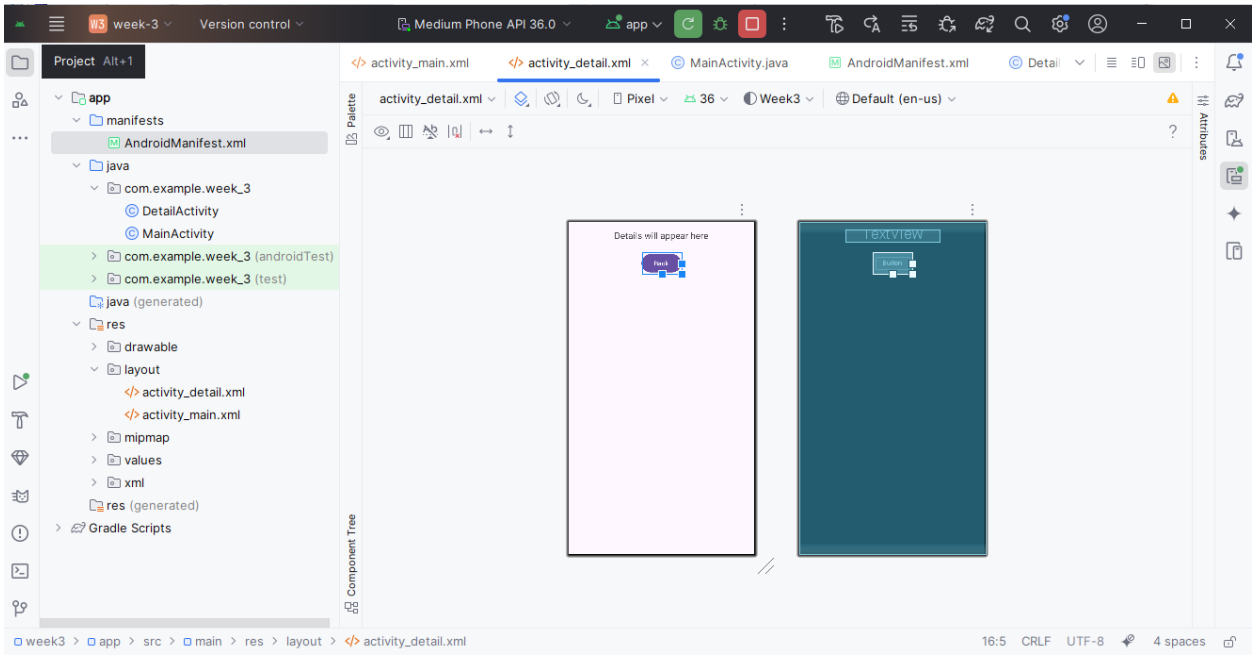
activity_detail.xml

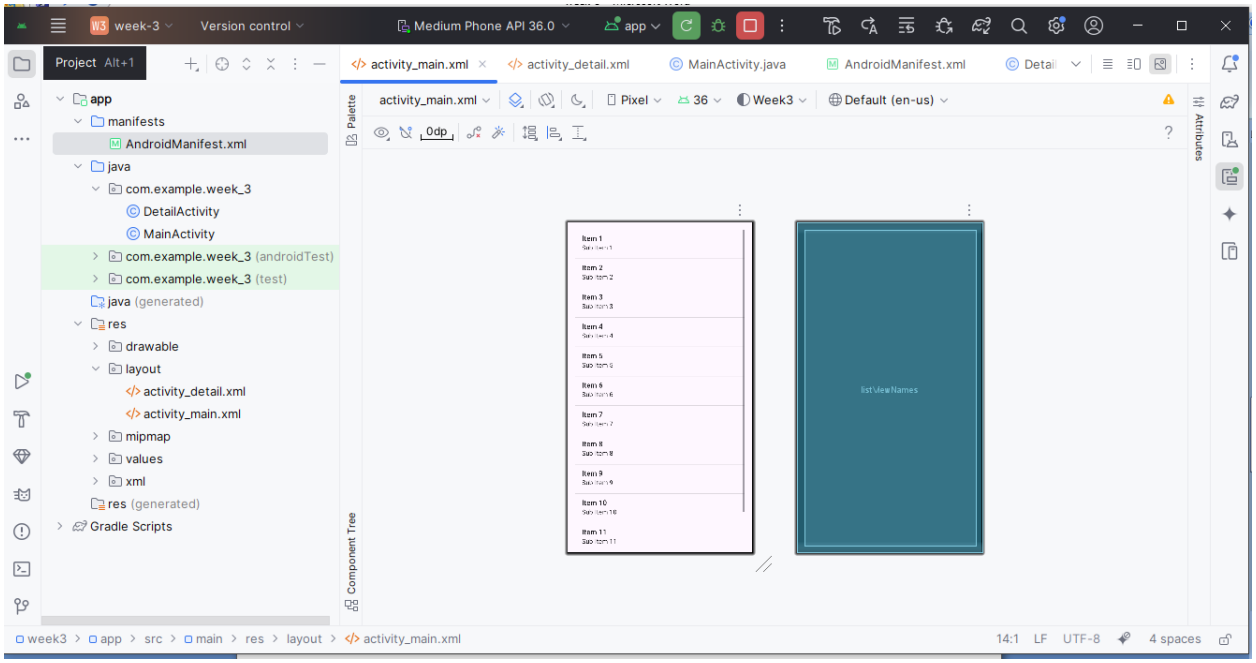
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:gravity="center_horizontal"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/tv"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:textSize="20sp"
      android:text="Details will appear here" />

   <Button
      android:id="@+id/b"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Back"
      android:layout_marginTop="24dp" />

</LinearLayout>
```





MainActivity.java

```
java
CopyEdit
package com.example.nameapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;

public class MainActivity extends AppCompatActivity {

    ListView lv;
    String[] nm = {"Alice", "Bob", "Charlie", "David"};
    String[] dt = {
        "Alice - Android Developer from NYC",
        "Bob - Data Scientist from LA",
        "Charlie - UX Designer from Chicago",
        "David - QA Engineer from Seattle"
    };

    @Override
    protected void onCreate(Bundle bd) {
        super.onCreate(bd);
        setContentView(R.layout.activity_main);

        lv = findViewById(R.id.lv);
        ArrayAdapter<String> ad = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, nm);
        lv.setAdapter(ad);

        lv.setOnItemClickListener((p, v, pos, id) -> {
            Intent i = new Intent(MainActivity.this, DetailActivity.class);
            i.putExtra("details", dt[pos]);
            startActivity(i);
        });
    }
}
```

DetailActivity.java

```
java
CopyEdit
package com.example.nameapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

public class DetailActivity extends AppCompatActivity {

    TextView tv;
    Button b;
```

```
    @Override
    protected void onCreate(Bundle bd) {
        super.onCreate(bd);
        setContentView(R.layout.activity_detail);

        tv = findViewById(R.id.tv);
        b = findViewById(R.id.b);

        // Get passed details from Intent
        String d = getIntent().getStringExtra("details");
        tv.setText(d);

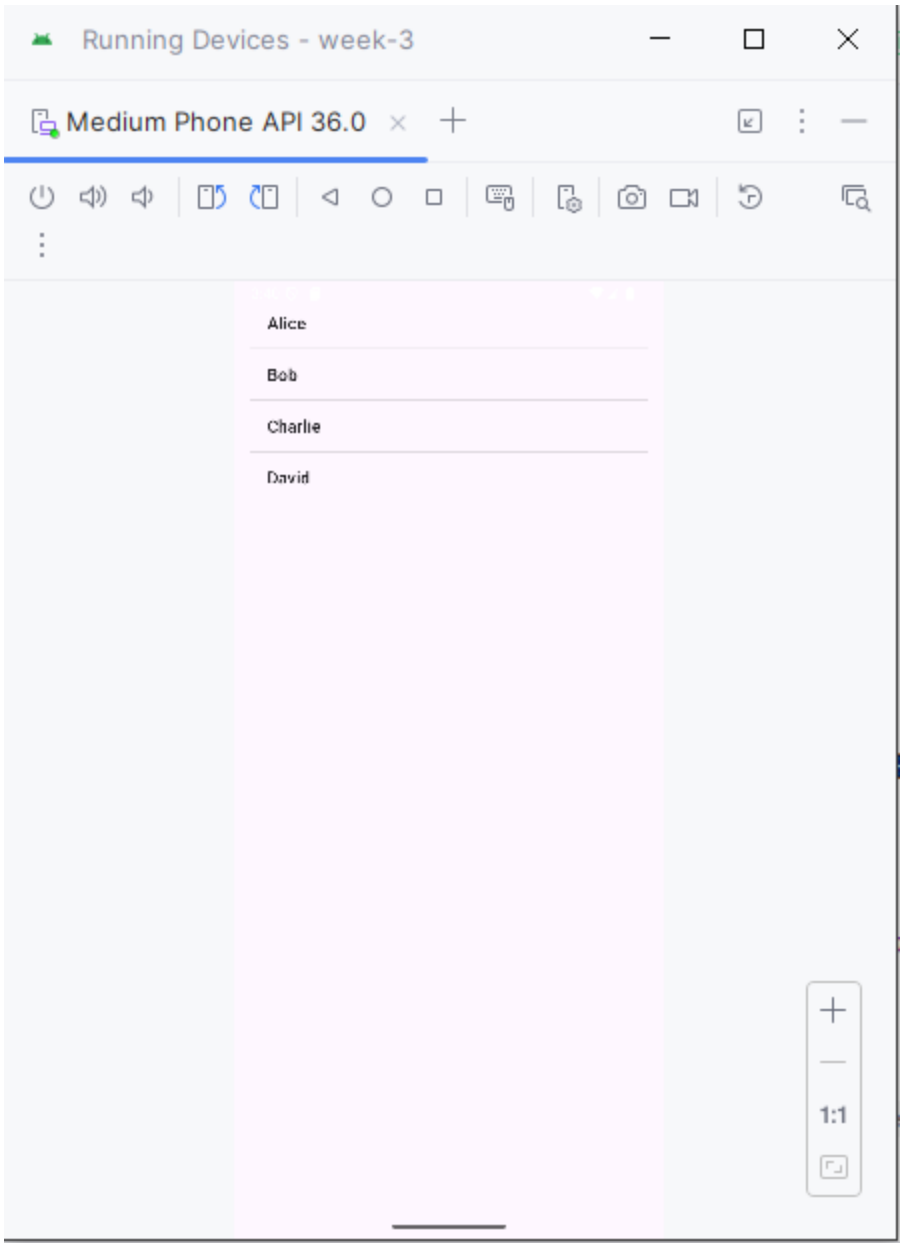
        b.setOnClickListener(v -> finish()); // Go back to previous screen
    }
}
```

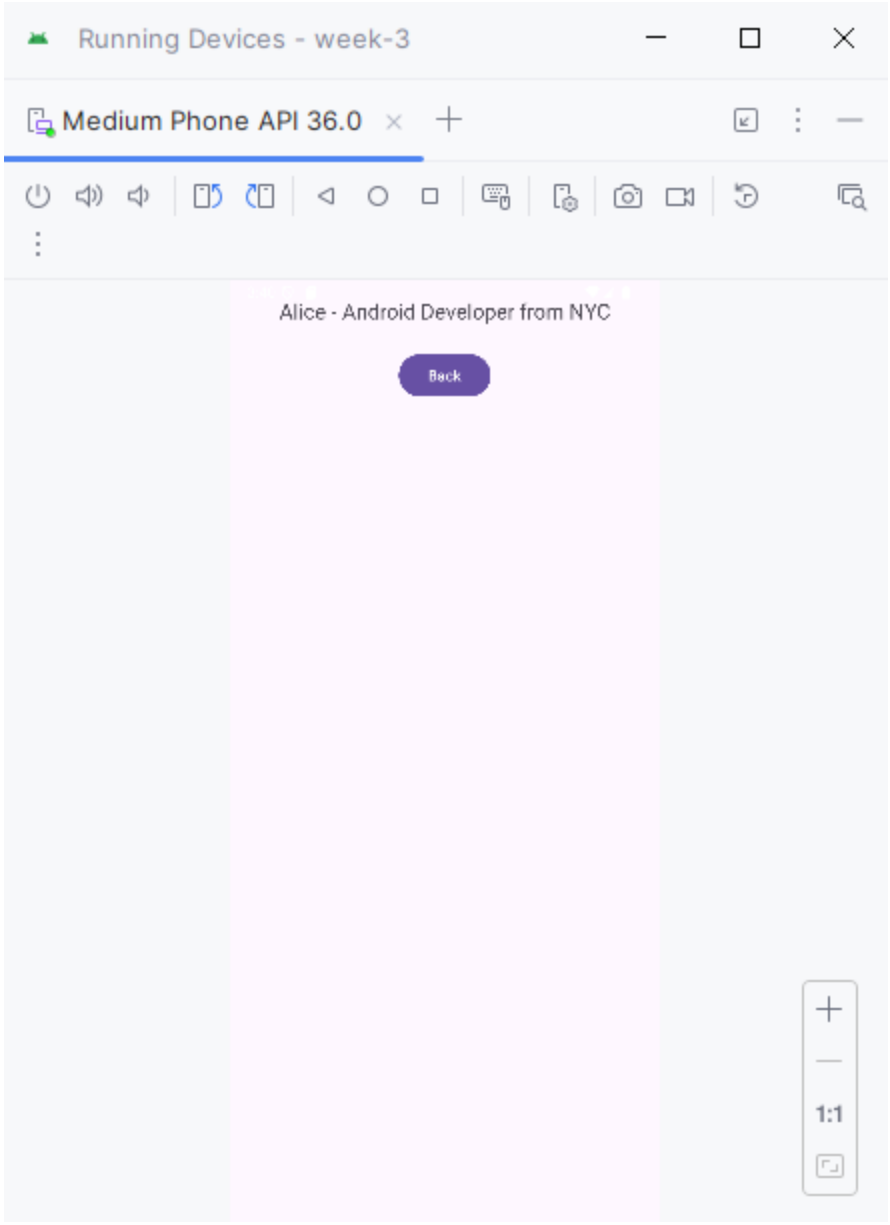
AndroidManifest.xml

```
xml
CopyEdit
<application
    ... >

    <activity android:name=".DetailActivity" />
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>

</application>
```


Running Devices - week-3
Medium Phone API 36.0
Alice
Bob
Charlie
David
1:1


Running Devices - week-3
Medium Phone API 36.0
Alice - Android Developer from NYC
Back
1:1

Week-4

Create a native calculator application.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   android:gravity="center">

   <EditText
      android:id="@+id/e1"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter first number"
      android:inputType="numberDecimal" />

   <EditText
      android:id="@+id/e2"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter second number"
      android:inputType="numberDecimal"
      android:layout_marginTop="8dp" />

   <LinearLayout
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
```

```
        android:layout_marginTop="16dp"
        android:gravity="center">

        <Button
            android:id="@+id/b1"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="+" />

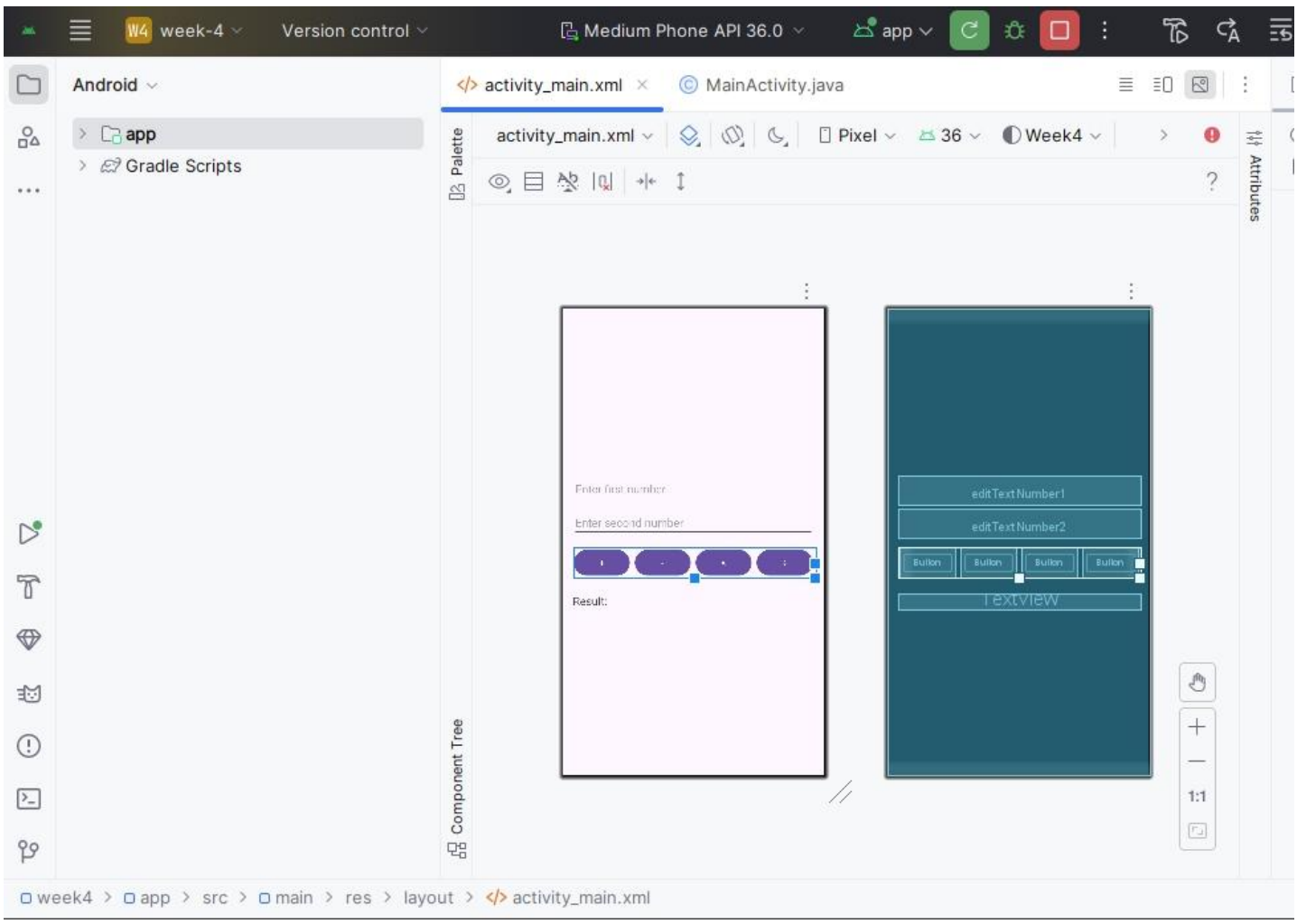
        <Button
            android:id="@+id/b2"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="-"
            android:layout_marginStart="8dp" />

        <Button
            android:id="@+id/b3"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="×"
            android:layout_marginStart="8dp" />

        <Button
            android:id="@+id/b4"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="÷"
            android:layout_marginStart="8dp" />
    </LinearLayout>
```

```
   <TextView
       android:id="@+id/tv"
       android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Result: "
       android:textSize="18sp"
       android:layout_marginTop="24dp" />

</LinearLayout>
```



MainActivity.java

```
package com.example.calculatorapp;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

   EditText e1, e2; Button b1, b2, b3, b4; TextView tv;

   @Override
   protected void onCreate(Bundle bd) {
super.onCreate(bd);
      setContentView(R.layout.activity_main);

      e1 = findViewById(R.id.e1); e2 = findViewById(R.id.e2); b1 =
      findViewById(R.id.b1);
      b2 = findViewById(R.id.b2); b3 =
      findViewById(R.id.b3);
      b4 = findViewById(R.id.b4);
      tv = findViewById(R.id.tv);

      b1.setOnClickListener(v -> calc('+'));
      b2.setOnClickListener(v -> calc('-'));
b3.setOnClickListener(v -> calc('*'));
b4.setOnClickListener(v -> calc('/'));
   }

   private void calc(char op) {
      String s1 = e1.getText().toString(); String s2 =
      e2.getText().toString();
```

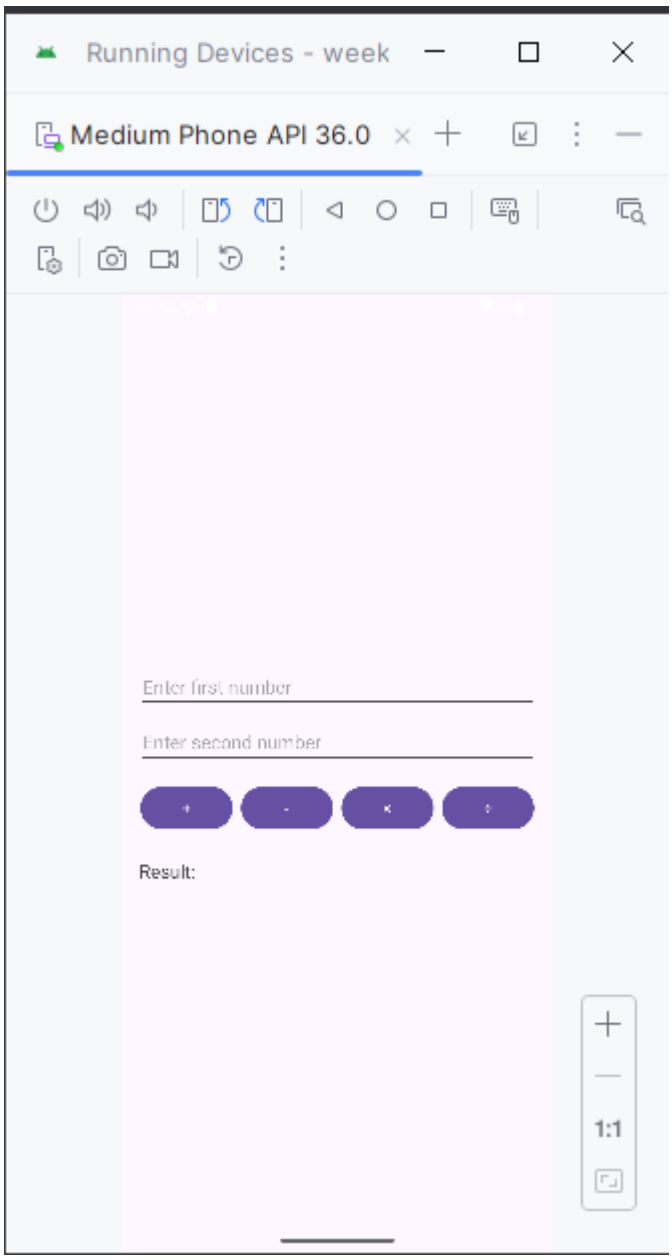
```
        if (s1.isEmpty() || s2.isEmpty()) {
            tv.setText("Please enter both numbers"); return;
        }

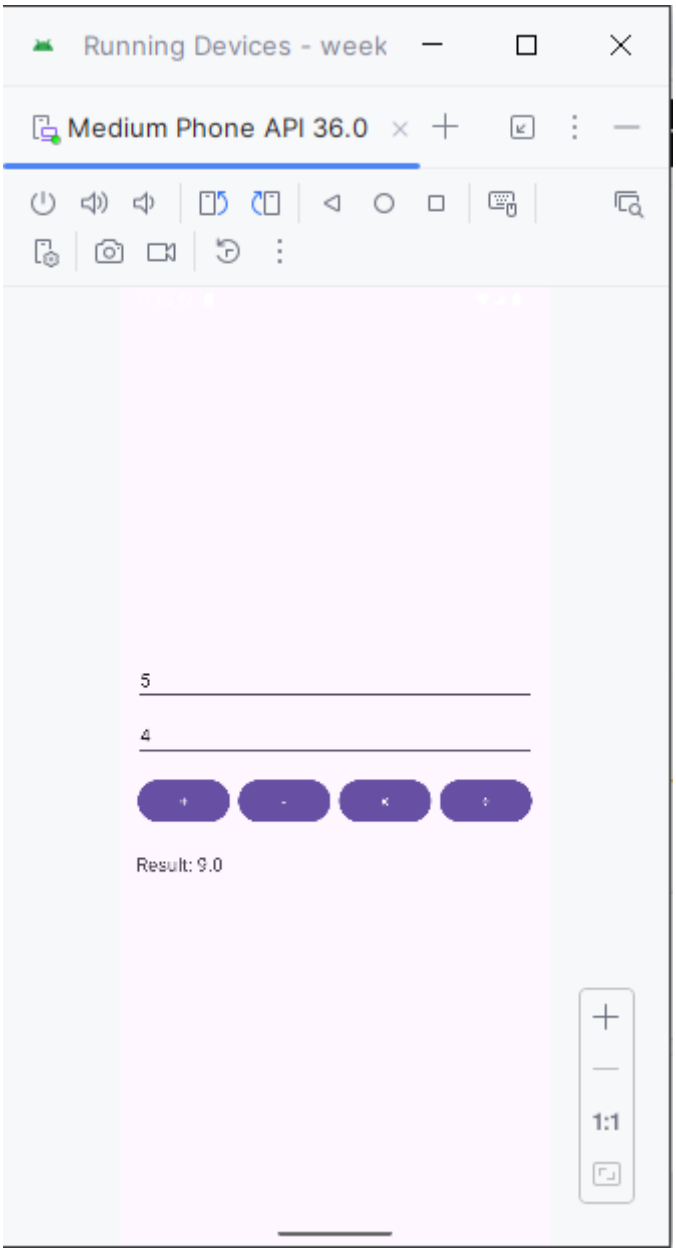
        double n1 = Double.parseDouble(s1); double n2 =
        Double.parseDouble(s2); double res = 0;

        switch (op) {
            case '+':
                res = n1 + n2;
                break;
            case '-':
                res = n1 - n2;
                break;
            case '*':
                res = n1 * n2;
                break;
            case '/':
                if (n2 == 0) {
                    tv.setText("Cannot divide by zero"); return;
                }
                res = n1 / n2;
                break;
        }

        tv.setText("Result: " + res); }
}
```

Output:


Running Devices - week
Medium Phone API 36.0
Enter first number
Enter second number
+
-
×
÷
Result:
1:1


Running Devices - week
Medium Phone API 36.0
5
4
+
-
×
÷
Result: 9.0
1:1

Week-5

Create an application that saves user information like name, age, gender etc. in shared preference and retrieves them when the program restarts.

Functionality

1. User enters `name`, `age`, and `gender`.
2. Info is saved using **SharedPreferences**.
3. When the app restarts, data is automatically loaded from preferences.

activity_main.xml (UI Layout)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">


   <EditText

      android:id="@+id/n"

      android:layout_width="match_parent"

android:layout_height="wrap_content"

android:hint="Name" />


   <EditText

      android:id="@+id/a"

      android:layout_width="match_parent"

android:layout_height="wrap_content"

android:inputType="number"

      android:hint="Age" />


   <EditText

      android:id="@+id/g"

      android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Gender" />

    <Button
        android:id="@+id/b"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Info" />

    <TextView
        android:id="@+id/tv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="20dp"
        android:text="User Info Will Be Displayed Here" />
</LinearLayout>
```

MainActivity.java

```
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText n, a, g; Button b;
    TextView tv;

    SharedPreferences sp;
```

```
    public static final String PN = "UserInfo";

    @Override
    protected void onCreate(Bundle bd) {
super.onCreate(bd);
        setContentView(R.layout.activity_main);

        n = findViewById(R.id.n);
        a = findViewById(R.id.a);
        g = findViewById(R.id.g); b =
findViewById(R.id.b);
        tv = findViewById(R.id.tv);

        sp = getSharedPreferences(PN, MODE_PRIVATE); load();

        b.setOnClickListener(v -> { save();
        });
    }

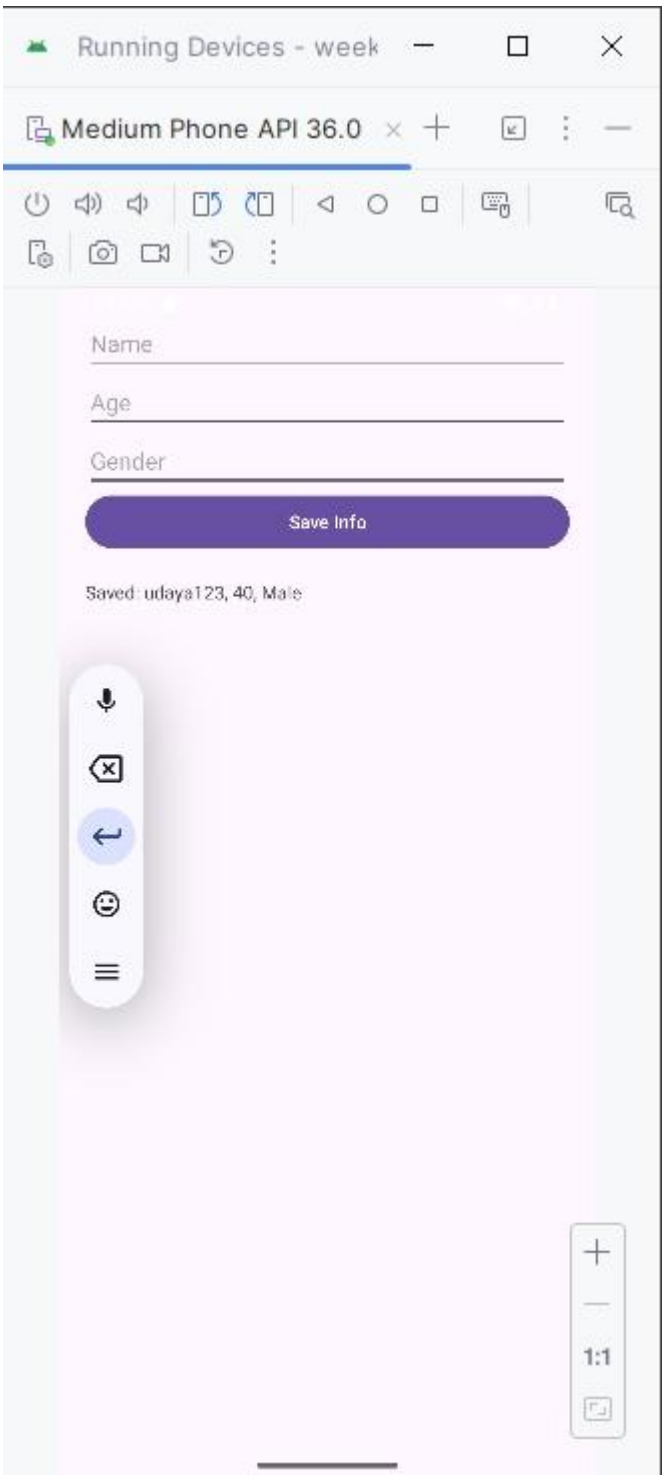
    private void save() {
        String nm = n.getText().toString();
        String ag = a.getText().toString();
        String gen = g.getText().toString();

        SharedPreferences.Editor ed = sp.edit();
ed.putString("Name", nm);
        ed.putString("Age", ag);
        ed.putString("Gender", gen);
        ed.apply();

        tv.setText("Saved: " + nm + ", " + ag + ", " + gen); }
```

```
    private void load() {
        String nm = sp.getString("Name", "Not set");
        String ag = sp.getString("Age", "Not set");
        String gen = sp.getString("Gender", "Not set");

        tv.setText("Loaded: " + nm + ", " + ag + ", " + gen); }
}
```


Running Devices - week
Medium Phone API 36.0
Name
Age
Gender
Save Info
Saved: uday123, 40, Male
1:1

Week-6

Develop a Registration and Login application that makes use of database.

The SQlite database as its name suggests, is a lightweight database which is used to store the data locally in your Android mobile devices. One can create a simple login and register app using SQlite which can also work as test bench or platform for launching the app for web servers.

Let us first create a Database helper class which extends the properties of SQliteOpenHelperClass.

The code snippet for DB Helper class "DBHelper.class" is as follows:

```
package com.learnandroid.loginsqlite;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import androidx.annotation.Nullable;

public class DBHelper  extends SQLiteOpenHelper {
public static final String DBN = "Login.db"; public
DBHelper(Context c) {
        super(c, "Login.db", null, 1); }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("create Table users(username TEXT primary key, password
TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int i, int i1) {
db.execSQL("drop Table if exists users");
    }

    public Boolean ins(String u, String p){ SQLiteDatabase db =
this.getWritableDatabase();
        ContentValues cv= new ContentValues();
        cv.put("username", u);
        cv.put("password", p); long res = db.insert("users", null,
cv); if(res==-1) return false;
        else
```

```
        return true;
    }

    public Boolean chkU(String u) {
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cur = db.rawQuery("Select * from users where username = ?",
new String[]{u});
        if (cur.getCount() > 0)
            return true;
        else
        return false;
    }

    public Boolean chkLogin(String u, String p){
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cur = db.rawQuery("Select * from users where username = ? and
password = ?", new String[] {u,p});
        if(cur.getCount()>0)
            return true;
        else
        return false;
    }
}
```

Let us first create main_activity.xml which serves as the register page. This page will have three edit text boxes, username, password and retype password. The code snippet for main_activity.xml is as follows.

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="10dp"
   tools:context=".MainActivity">

    <EditText
    android:id="@+id/u"
    android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="User Name"
    android:layout_marginTop="50dp"/>

    <EditText
    android:id="@+id/p"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/u"/>

    <EditText
    android:id="@+id/rp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Retype Password"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/p"/>

    <Button
    android:id="@+id/su"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/rp"/>

    <Button
    android:id="@+id/si"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Existing user! Go to Sign in page"
android:layout_marginTop="50dp"
    android:layout_below="@+id/su"/>

</RelativeLayout>
```

The code snippet for MainActivity.java is as follows.

```
package com.learnandroid.loginsqlite;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity  extends AppCompatActivity {

    EditText u, p, rp;
    Button su, si;
    DBHelper db;
```

```
    @Override
    protected void onCreate(Bundle bd) {
super.onCreate(bd);
        setContentView(R.layout.activity_main);

        u = (EditText) findViewById(R.id.u);
        p = (EditText) findViewById(R.id.p);
        rp = (EditText) findViewById(R.id.rp); su = (Button)
findViewById(R.id.su);
        si = (Button) findViewById(R.id.si);
        db = new DBHelper(this);

        su.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View v) {
                String us = u.getText().toString();
                String pw = p.getText().toString();
                String rpw = rp.getText().toString();

                if(us.equals("")||pw.equals("")||rpw.equals(""))
Toast.makeText(MainActivity.this, "Please enter all the fields",
Toast.LENGTH_SHORT).show();
                else{
                    if(pw.equals(rpw)){
                        Boolean cu = db.chkU(us);
                        if(cu==false){
                            Boolean ins = db.ins(us, pw);
                            if(ins==true){
                                Toast.makeText(MainActivity.this,
"Registered successfully", Toast.LENGTH_SHORT).show();
                                Intent i = new
Intent(getApplicationContext(),HomeActivity.class);
                                startActivity(i);
                            }else{
                                Toast.makeText(MainActivity.this,
"Registration failed", Toast.LENGTH_SHORT).show();

                            }
                        }
                        else{
                            Toast.makeText(MainActivity.this, "User
already exists! please sign in", Toast.LENGTH_SHORT).show();
                        }
                    }else{
                        Toast.makeText(MainActivity.this, "Passwords not
matching", Toast.LENGTH_SHORT).show();
                    }
```

```
                } }
        });

        si.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent i = new Intent(getApplicationContext(),
LoginActivity.class);
                startActivity(i);
            }
        });
    }
}
```

Now create another activity for Login. The code snippet for activity_login.xml is as follows.

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="10dp"
   tools:context=".LoginActivity">

      <EditText
      android:id="@+id/u1"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="User Name"
      android:layout_marginTop="50dp"/>

      <EditText
      android:id="@+id/p1"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Password"
      android:layout_marginTop="50dp"

      android:layout_below="@+id/u1"/>

      <Button
      android:id="@+id/b"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Sign in"
      android:layout_marginTop="50dp"
      android:layout_below="@+id/p1"/>
</RelativeLayout>
```

The code snippet for LoginActivity.java is as follows:

```
package com.learnandroid.loginsqlite;
import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.Toast;

public class LoginActivity extends AppCompatActivity {
    EditText u, p;
    Button b;
    DBHelper db;
    @Override
    protected void onCreate(Bundle bd) {
super.onCreate(bd);
        setContentView(R.layout.activity_login);

        u = (EditText) findViewById(R.id.u1); p = (EditText)
findViewById(R.id.p1); b = (Button) findViewById(R.id.b); db
= new DBHelper(this);

        b.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View v) {

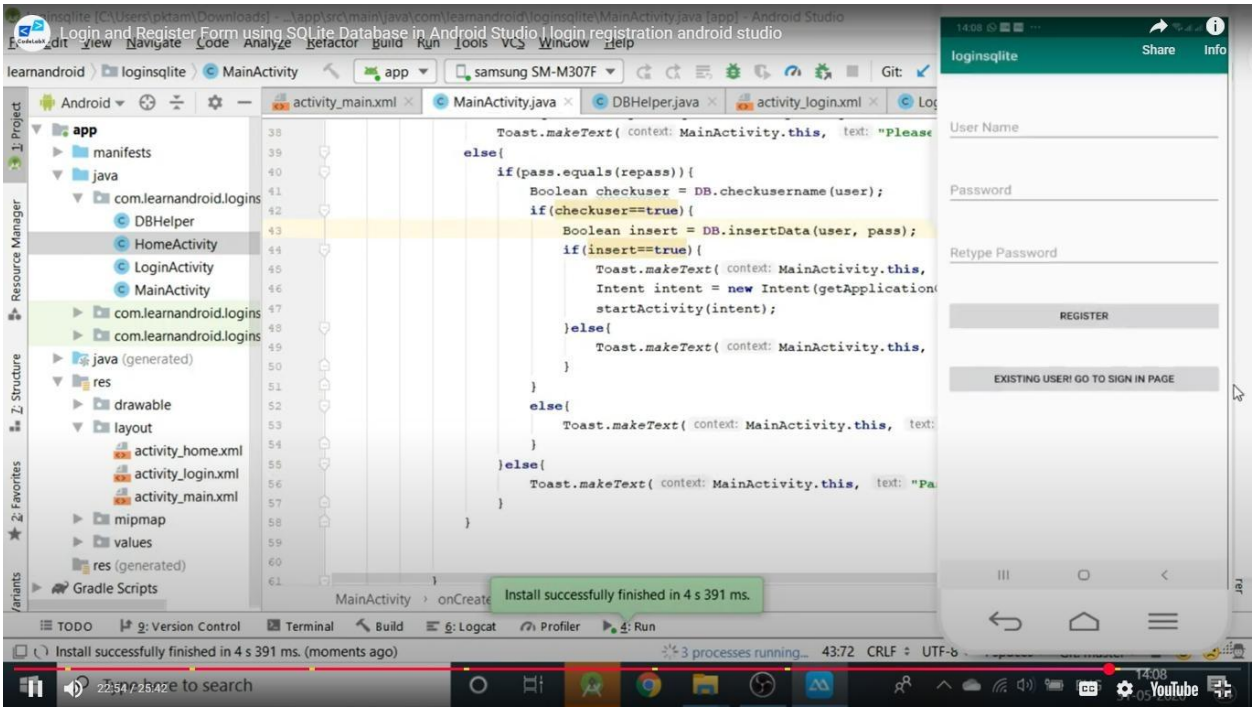
                String us = u.getText().toString(); String pw =
p.getText().toString();

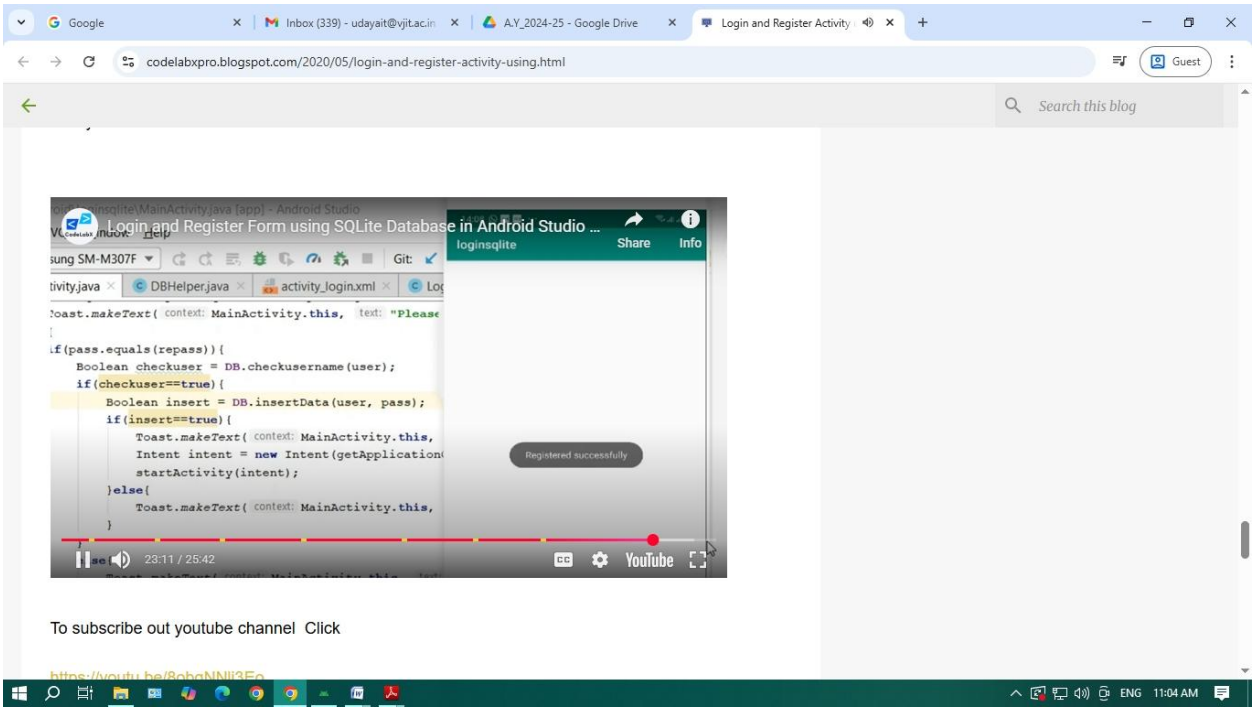
                if(us.equals("")||pw.equals(""))

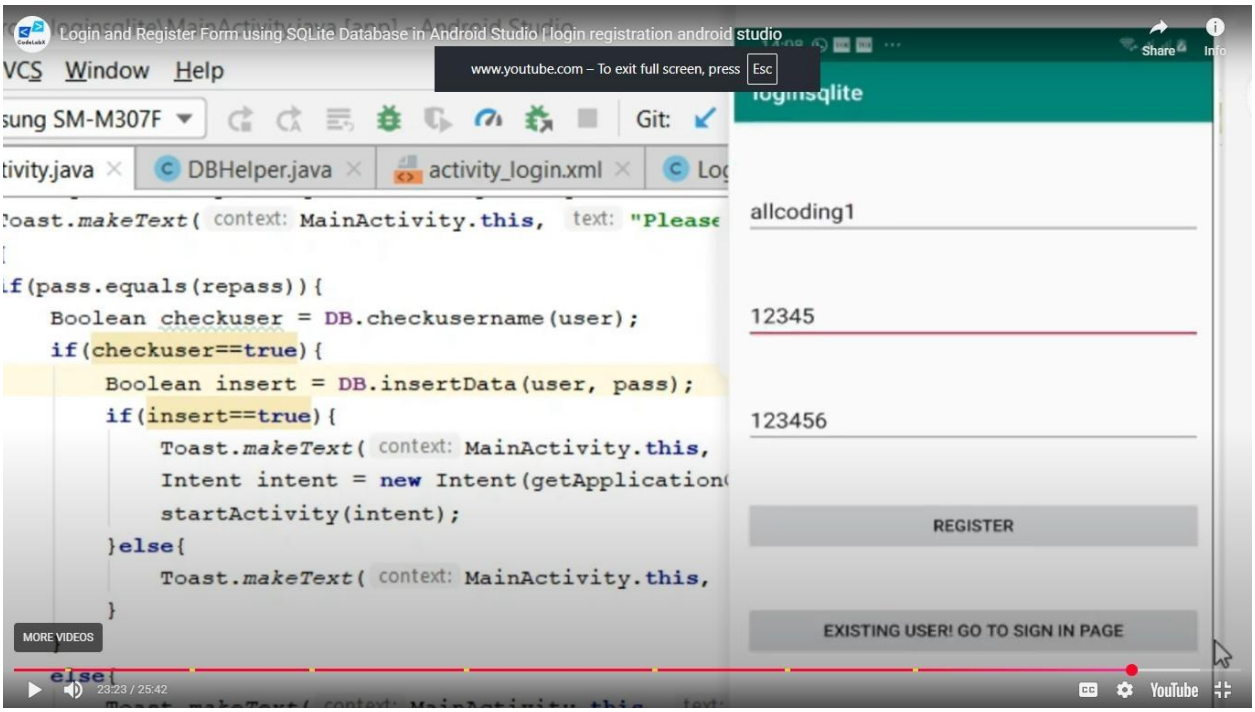
                    Toast.makeText(LoginActivity.this, "Please enter all the
fields", Toast.LENGTH_SHORT).show();
                else{
                    Boolean ok =
db.chkLogin(us, pw);
                    if(ok==true){
                        Toast.makeText(LoginActivity.this, "Sign in
successfull", Toast.LENGTH_SHORT).show();
                        Intent i = new
Intent(getApplicationContext(), HomeActivity.class);
                        startActivity(i);
                    }else{
                        Toast.makeText(LoginActivity.this, "Invalid
Credentials", Toast.LENGTH_SHORT).show();
                    }
```

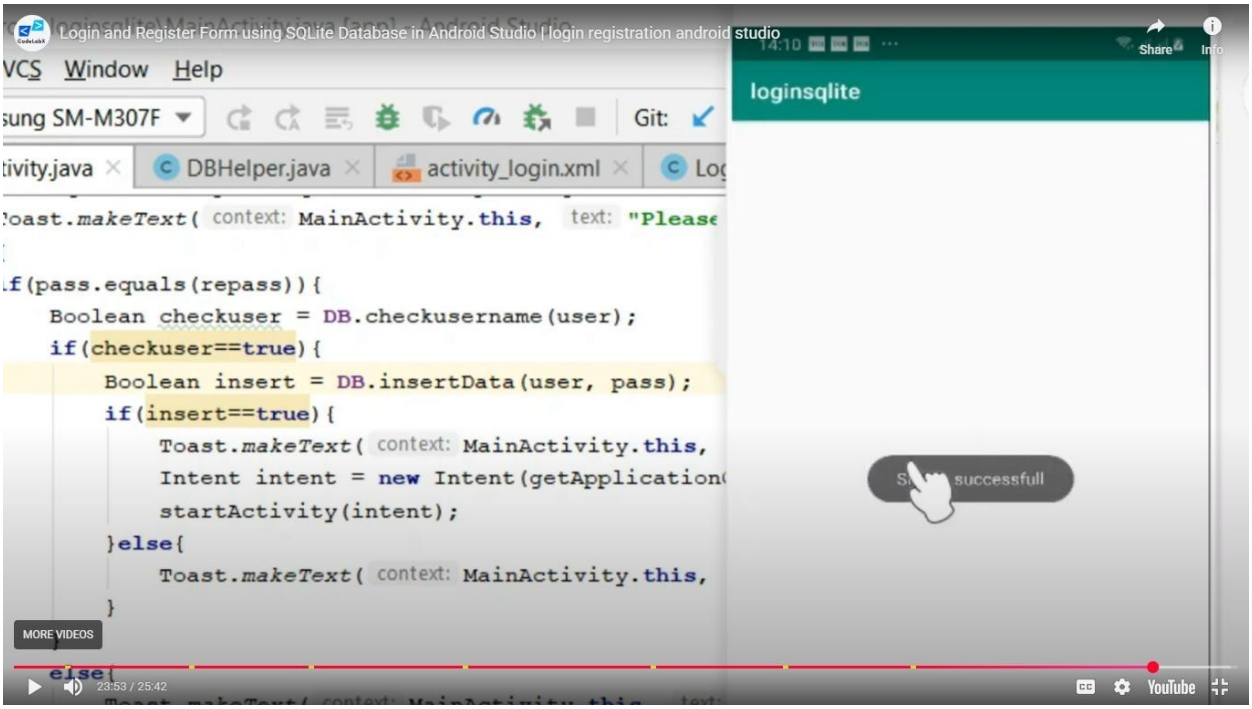
```
                }
            }
        });
    }
}
```

Output:


Login and Register Form using SQLite Database in Android Studio | login registration android studio
loginsqlite
Share
Info
User Name
Password
Retype Password
REGISTER
EXISTING USER! GO TO SIGN IN PAGE
Install successfully finished in 4 s 391 ms.
Install successfully finished in 4 s 391 ms. (moments ago)


Google
Inbox (339) - udayait@vjit.ac.in
A.Y_2024-25 - Google Drive
Login and Register Activity
codelabxpro.blogspot.com/2020/05/login-and-register-activity-using.html
Guest
Search this blog
Login and Register Form using SQLite Database in Android Studio ...
loginsqlite
Share
Info
Registered successfully
23:11 / 25:42
YouTube
To subscribe out youtube channel Click
ENG
11:04 AM


Login and Register Form using SQLite Database in Android Studio | login registration android studio
www.youtube.com – To exit full screen, press Esc
Share
Info
VCS Window Help
loginsqlite
allcoding1
12345
123456
REGISTER
EXISTING USER! GO TO SIGN IN PAGE
MORE VIDEOS
23:23 / 25:42
YouTube



Supporting link is “https://codelabxpro.blogspot.com/2020/05/login-and-register-activity-using.html”

Week 7:

Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Activtity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <Button
        android:id="@+id/b"
android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Create Notification"
        tools:layout_editor_absoluteX="170dp"
        tools:layout_editor_absoluteY="322dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Mainactivity.java

```
package com.example.notify;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

import android.annotation.SuppressLint;
import android.os.Bundle;

/*public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}*/
import android.app.NotificationChannel ;
import android.app.NotificationManager ;
import android.content.ContentResolver ;
import android.content.Context ;
import android.graphics.Color ;
import android.media.AudioAttributes ;
import android.net.Uri ;
//import android.support.v4.app.NotificationCompat ;
//import android.support.v7.app.AppCompatActivity ;
//import android.os.Bundle ;
import android.view.View ;
```

```
import android.widget.Button ;

import java.util.Objects;

public class MainActivity extends AppCompatActivity {
    public static final String CID = "10001" ; private final static
String ch;

    static {
        ch = "default"; }

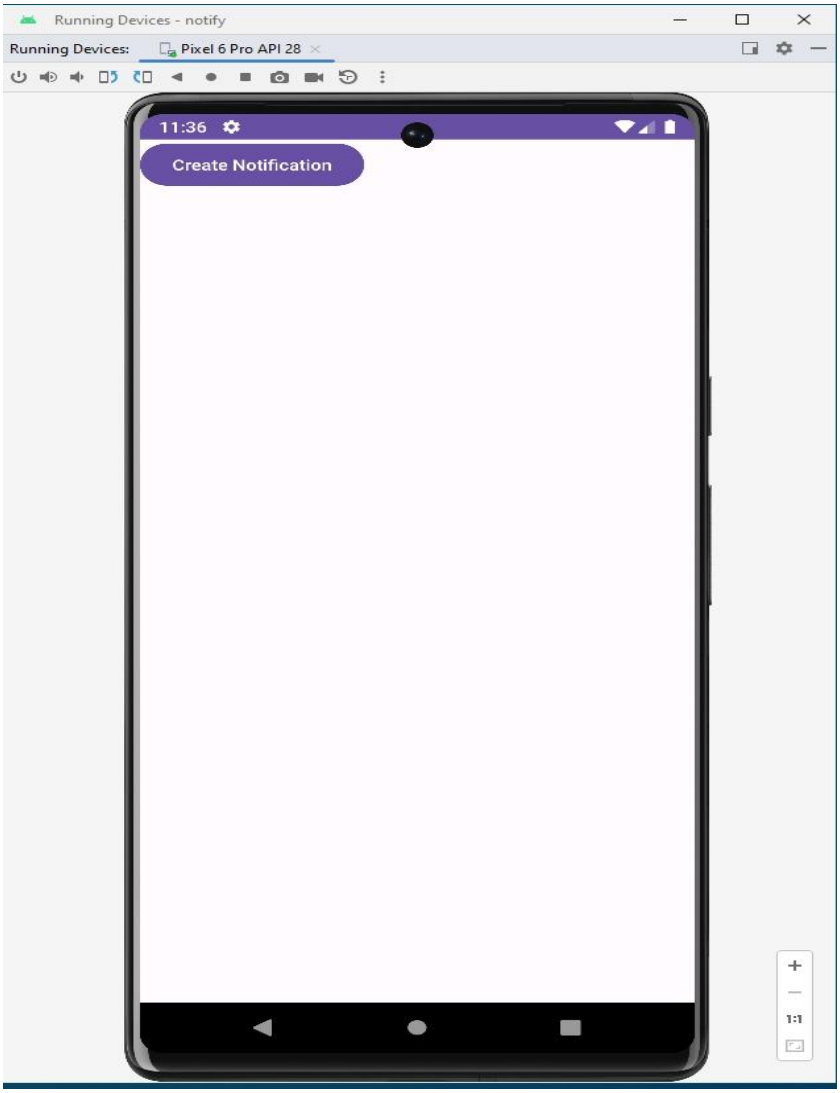
    @Override
    protected void onCreate (Bundle bd) {
        super.onCreate(bd) ;
        setContentView(R.layout.activity_main ) ;
        @SuppressLint({"MissingInflatedId", "LocalSuppress"}) Button b
= findViewById(R.id.b ) ;
        b.setOnClickListener( new View.OnClickListener() {
            @SuppressLint("NotificationPermission")
            @Override
            public void onClick (View v) {
                Uri s = Uri.parse
(ContentResolver.SCHEME_ANDROID_RESOURCE + "://" +
getPackageName() + "/raw/quite_impressed.mp3" ) ;
                NotificationCompat.Builder nb = new
NotificationCompat.Builder(MainActivity.this,
                        ch )
                        .setSmallIcon(R.drawable.ic_launcher_foreground )
                        .setContentTitle( "Test" )
                        .setSound(s)
                        .setContentText( "Hello! This is my first push
notification" );
                NotificationManager nm = (NotificationManager)
getSystemService(Context.NOTIFICATION_SERVICE ) ;
                if (android.os.Build.VERSION.SDK_INT >=
android.os.Build.VERSION_CODES.O ) {
                    AudioAttributes aa = new
AudioAttributes.Builder()

.setContentType(AudioAttributes.CONTENT_TYPE_SONIFICATION )
                            .setUsage(AudioAttributes.USAGE_ALARM )
                            .build() ;
                    int imp = NotificationManager.IMPORTANCE_HIGH ;
NotificationChannel nc = new
                            NotificationChannel( CID ,
"NOTIFICATION_CHANNEL_NAME" , imp) ;
                    nc.enableLights( true ) ;
                    nc.setLightColor(Color.RED ) ;
                    nc.enableVibration( true ) ;
                    nc.setVibrationPattern( new long []{ 100 , 200 , 300 , 400
, 500 , 400 , 300 , 200 , 400 }) ;
                    nc.setSound(s , aa) ; nb.setChannelId( CID ) ;
                    assert nm != null;

nm.createNotificationChannel(nc) ;
                }
                Objects.requireNonNull(nm).notify(( int )
System.currentTimeMillis (), nb.build()) ;
```

```
                }
            }) ;
        }
    }
```

output:


Running Devices - notify
Running Devices:
Pixel 6 Pro API 28
11:36
Create Notification


Running Devices - notify
Running Devices:
Pixel 6 Pro API 28
notify
Test
Hello! This is my first push notification

Week-8

Develop an application with three options: dial a phone number, open a website, and send an SMS. Each option uses an Intent.

Step-by-Step Implementation

1. Open Android Studio and create an Empty Views Activity.
2. Enter the project name: IntentMenuApp.
3. Select Java and choose Minimum SDK: API 21 or above.

activity_main.xml

This layout displays the same three buttons: Dial a Number, Open a Website, and Send an SMS.

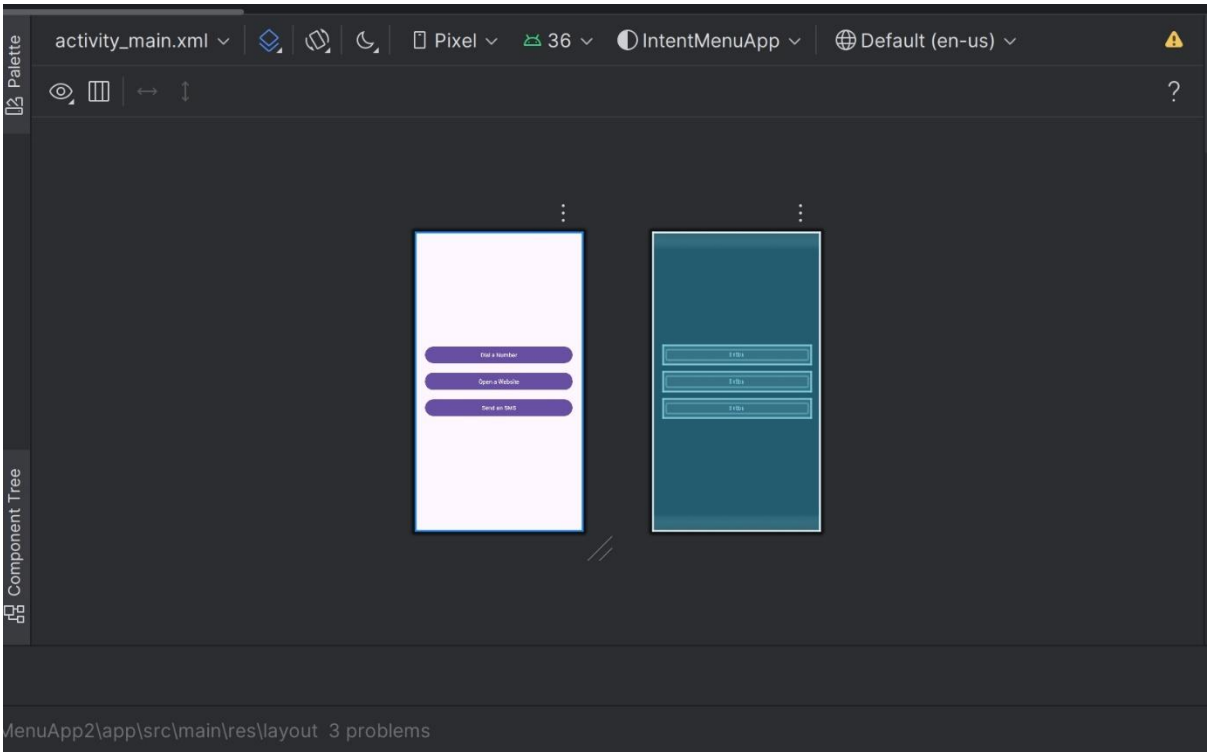
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <Button
        android:id="@+id/d"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Dial a Number" />

    <Button
        android:id="@+id/w"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Open a Website"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/s"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send an SMS"
        android:layout_marginTop="16dp" />

</LinearLayout>
```



MainActivity.java

The three short click listeners open the dialer, browser, and SMS app.

```
package com.example.intentmenuapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    Button d, w, s;

    @Override
    protected void onCreate(Bundle bd) {
        super.onCreate(bd);
        setContentView(R.layout.activity_main);

        d = findViewById(R.id.d);
        w = findViewById(R.id.w);
        s = findViewById(R.id.s);

        d.setOnClickListener(v -> {
            Intent i = new Intent(Intent.ACTION_DIAL,
                    Uri.parse("tel:1234567890"));
            startActivity(i);
        });

        w.setOnClickListener(v -> {
            Intent i = new Intent(Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com"));
```

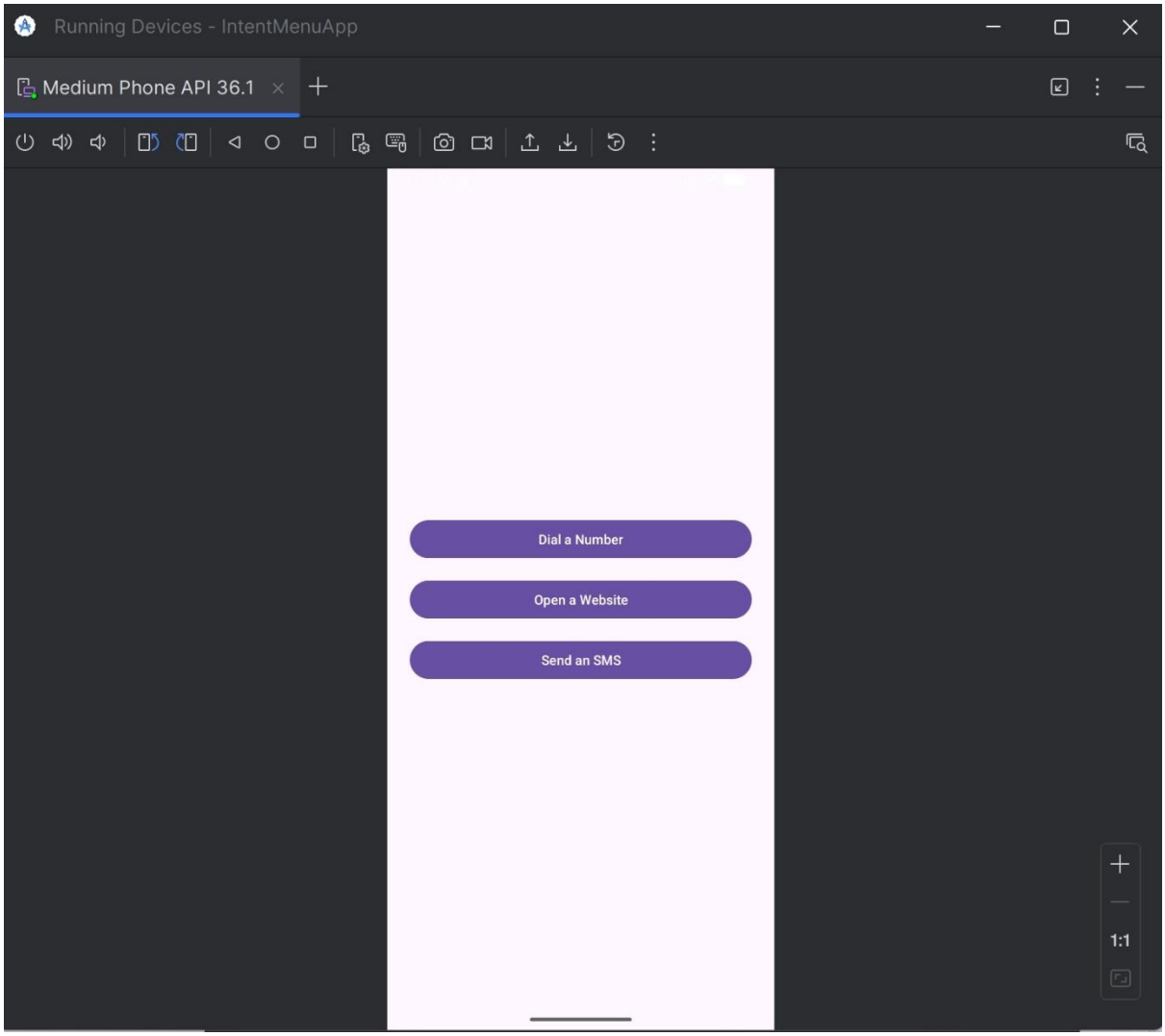
```
            startActivity(i);
        });

        s.setOnClickListener(v -> {
            Intent i = new Intent(Intent.ACTION_VIEW,
                    Uri.parse("sms:1234567890"));
            i.putExtra("sms_body", "Hello! This is a test message.");
            try {
                startActivity(i);
            } catch (Exception e) {
                Toast.makeText(this, "No SMS app found!",
                        Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

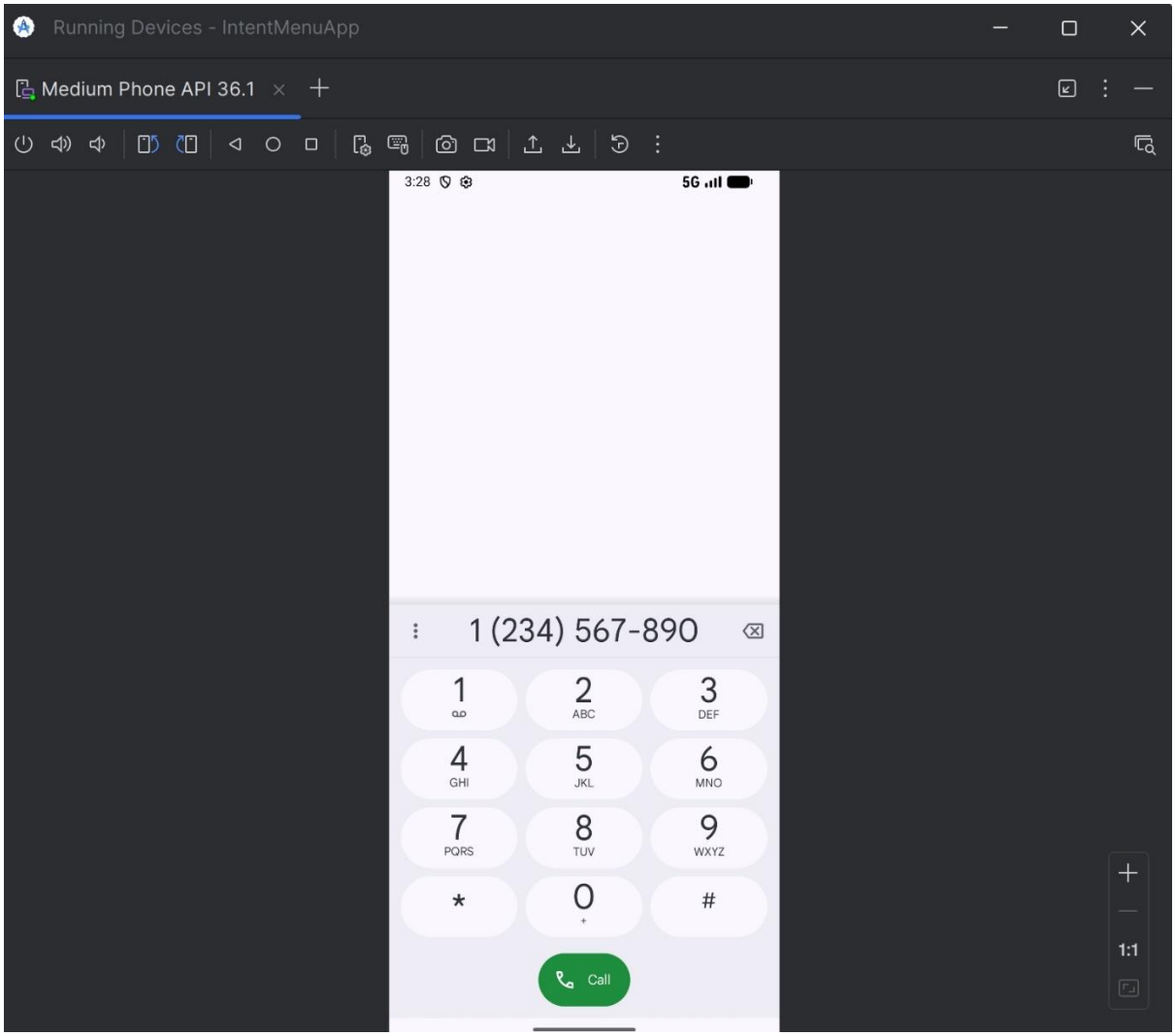
AndroidManifest.xml

No extra permissions are required. Keep the default AndroidManifest.xml created by Android Studio.

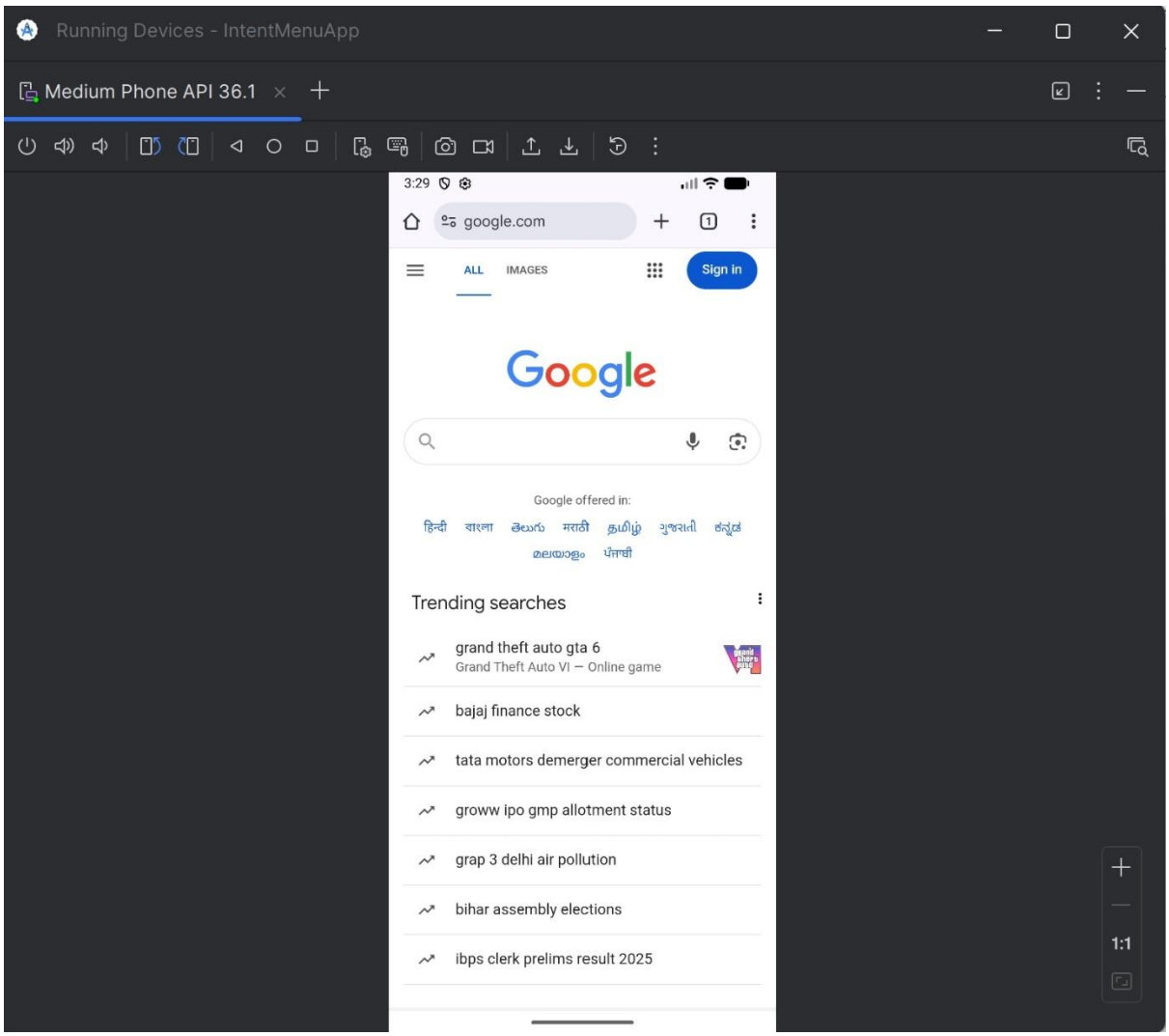
Output



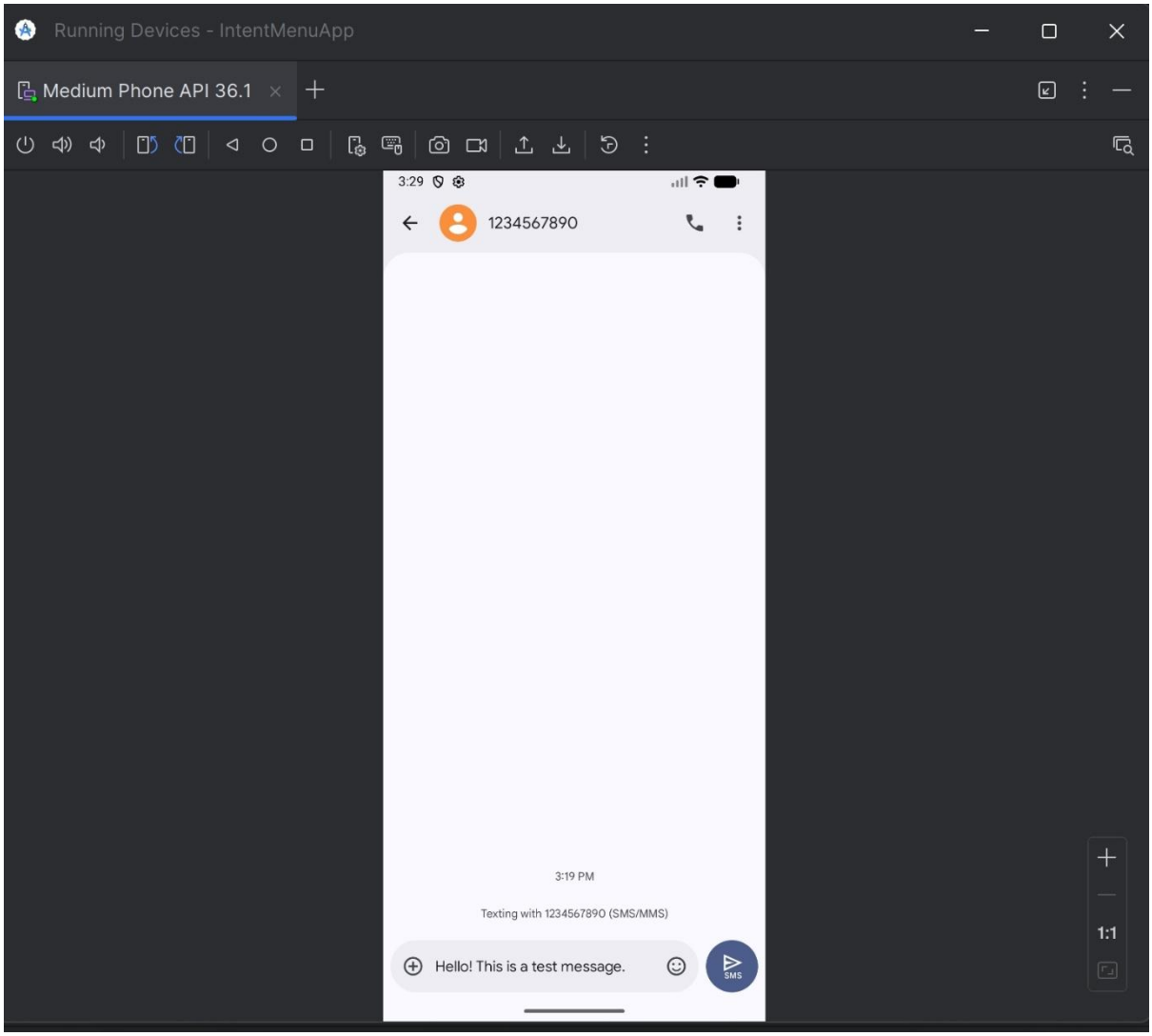
Main Screen: Displays all three buttons.



Dial Number: Opens the phone dialer with 1234567890.



Open Website: Opens https://www.google.com in the browser.



SMS: Opens the SMS app with the sample message already filled in.

Week-9

Create an alarm application using time picker to set alarm time.

Create the Layout (activity_main.xml)

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <Button
      android:id="@+id/b"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Set Alarm Time" />
```

```
   <TextView
      android:id="@+id/tv"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="No Alarm Set" />

</LinearLayout>
```

MainActivity.java

```
package com.example.alarmapp;

//package com.example.alarmapp;

import android.app.AlarmManager;
import android.app.PendingIntent;
import android.app.TimePickerDialog;
import android.content.Context; import
android.content.Intent;
import android.os.Bundle;

import android.widget.Button;
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
private TextView tv;
   private AlarmManager am;
   private PendingIntent pi;
   private Calendar cal = Calendar.getInstance();

   @Override
   protected void onCreate(Bundle bd) {
super.onCreate(bd);
      setContentView(R.layout.activity_main);

     tv = findViewById(R.id.tv);
     Button b = findViewById(R.id.b);
     am = (AlarmManager) getSystemService(Context.ALARM_SERVICE);
```

```
        b.setOnClickListener(v -> {
            int h = cal.get(Calendar.HOUR_OF_DAY); int m =
cal.get(Calendar.MINUTE);

            TimePickerDialog dlg = new TimePickerDialog(MainActivity.this, (TimePicker
tp, int sh, int sm) -> {
                        cal.set(Calendar.HOUR_OF_DAY, sh);
                        cal.set(Calendar.MINUTE, sm);
                        cal.set(Calendar.SECOND, 0);

                        tv.setText("Alarm set for: " + sh + ":" +
String.format("%02d", sm));
                        alarm(cal);
                    }, h, m, true);
            dlg.show();
        });
    }

    private void alarm(Calendar c) {

        Intent i = new Intent(this, AlarmReceiver.class);
        pi = PendingIntent.getBroadcast(this, 0, i,
PendingIntent.FLAG_IMMUTABLE);
        am.set(AlarmManager.RTC_WAKEUP, c.getTimeInMillis(), pi);

        Toast.makeText(this, "Alarm Scheduled",
Toast.LENGTH_SHORT).show(); }
}
```

AlarmReceiver.java

```
package com.example.alarmapp;

//package com.example.alarmapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class AlarmReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context ctx, Intent i) {
```

```
        Toast.makeText(ctx, "Alarm! Wake up!", Toast.LENGTH_LONG).show(); //
Can also trigger a notification here
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">

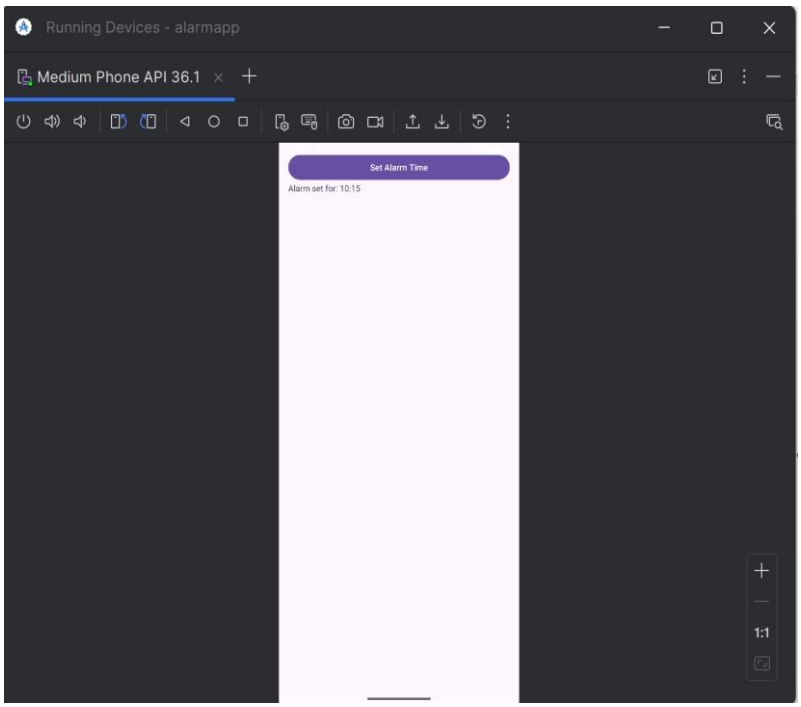
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"

        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Alarmapp">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

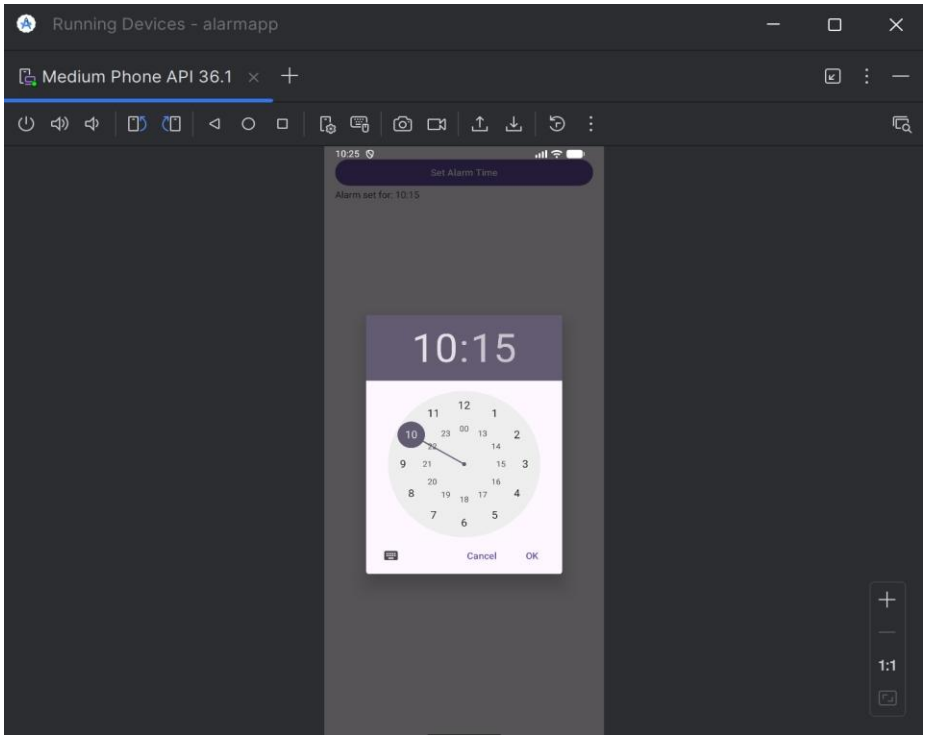
                <category android:name="android.intent.category.LAUNCHER"
/>            </intent -filter>
        </activity>
        <receiver android:name=".AlarmReceiver"/>
    </application>


</manifest>
```

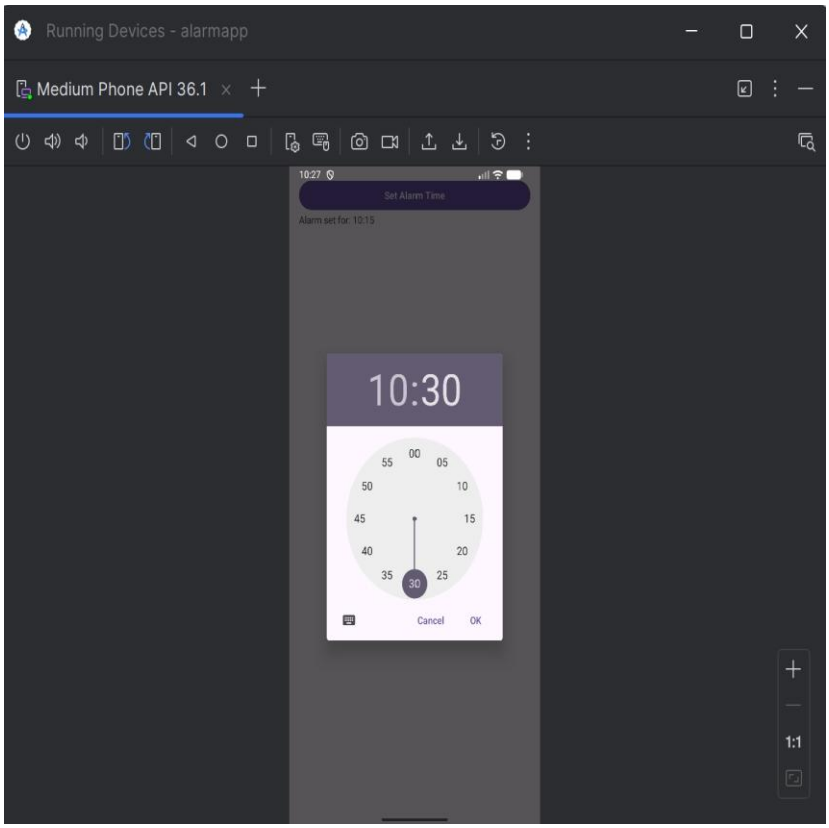
Output:

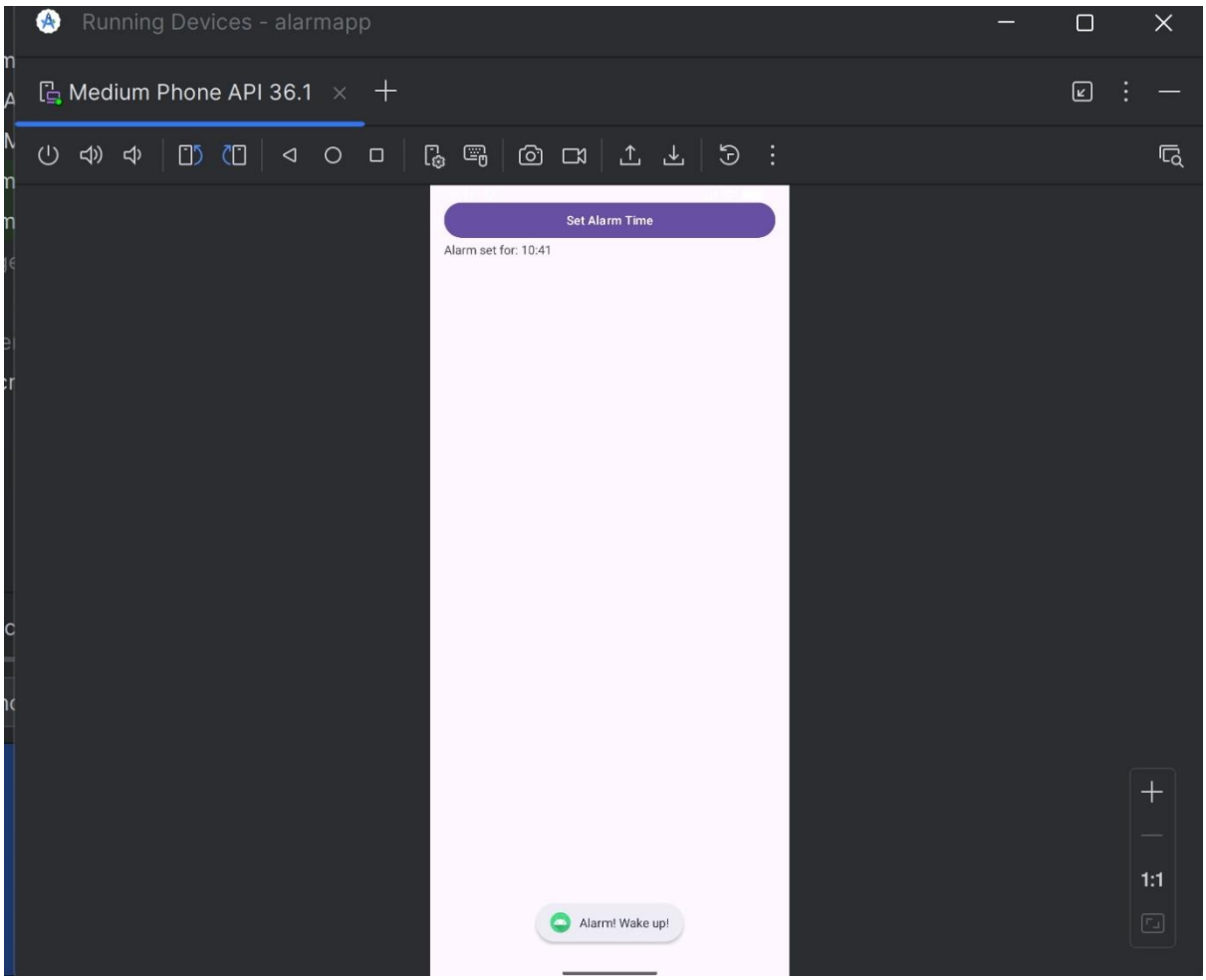

Running Devices - alarmapp
Medium Phone API 36.1
Set Alarm Time
Alarm set for: 10:15
1:1

Hours:



Minitues:




Running Devices - alarmapp
Medium Phone API 36.1
Set Alarm Time
Alarm set for: 10:41
Alarm! Wake up!
1:1

Week-10

Create an application that shows the given URL (from a text field) in a browser

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.openurlapp">

    <application
        android:allowBackup="true"
        android:label="Open URL App"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category
android:name="android.intent.category.LAUNCHER"/>            </intent -
filter>
        </activity>

    </application>

</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation ="vertical"
   android:padding="24dp"
   android:gravity="center">

   <EditText
      android:id="@+id/et"
      android:layout_width ="match_parent"
      android:layout_height ="wrap_content"
      android:hint="Enter URL (e.g. https://www.google.com)"
android:inputType="textUri"
      android:layout_marginBottom ="16dp" />

   <Button
      android:id="@+id/b"
      android:layout_width ="wrap_content"
android:layout_height ="wrap_content"
android:text="Open in Browser"
      android:backgroundTint="#4CAF50"
      android:textColor="#FFFFFF"
      android:padding="10dp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.openurlapp;


import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;
```

```
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   EditText et; Button b;

   @Override
   protected void onCreate(Bundle bd) {
super.onCreate(bd);
       setContentView(R.layout.activity_main);

      et = findViewById(R.id.et); b =
      findViewById(R.id.b);

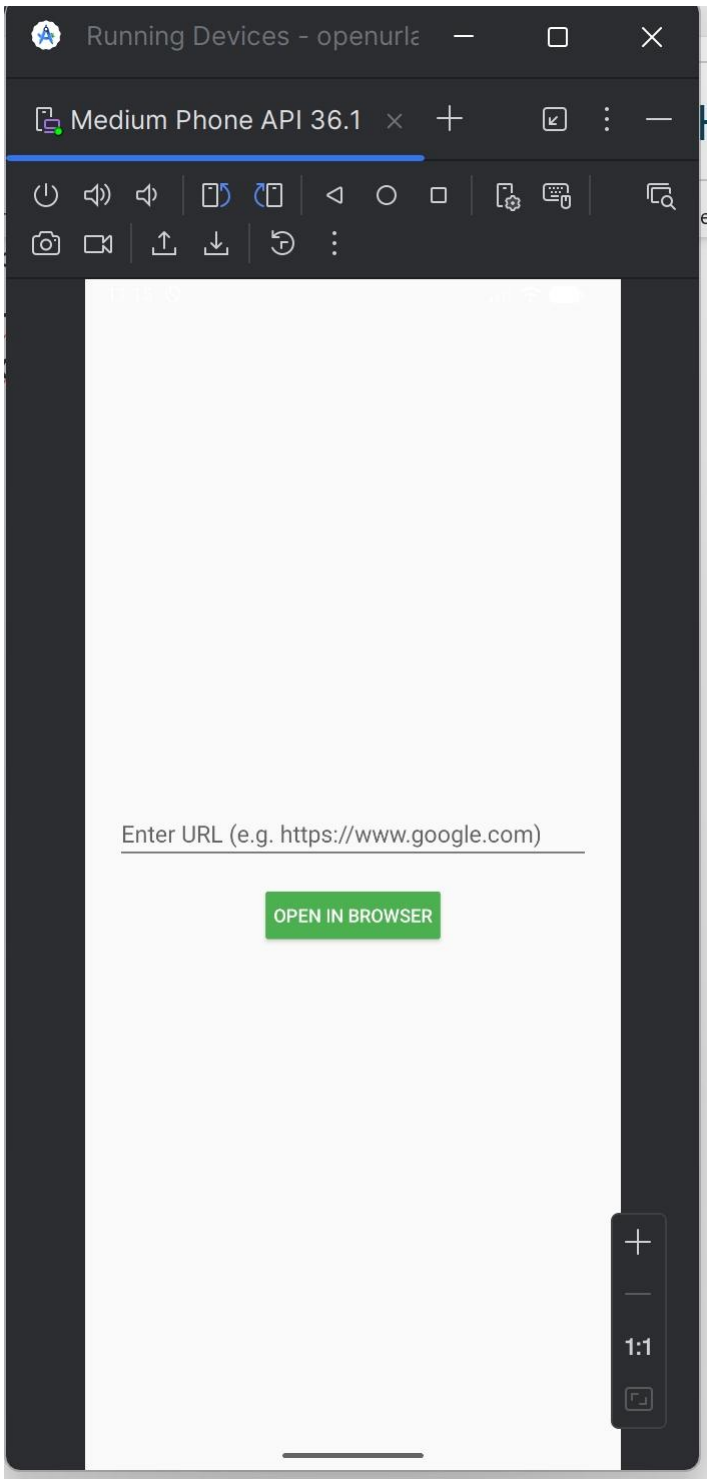
       b.setOnClickListener(v -> {
          String u = et.getText().toString().trim();

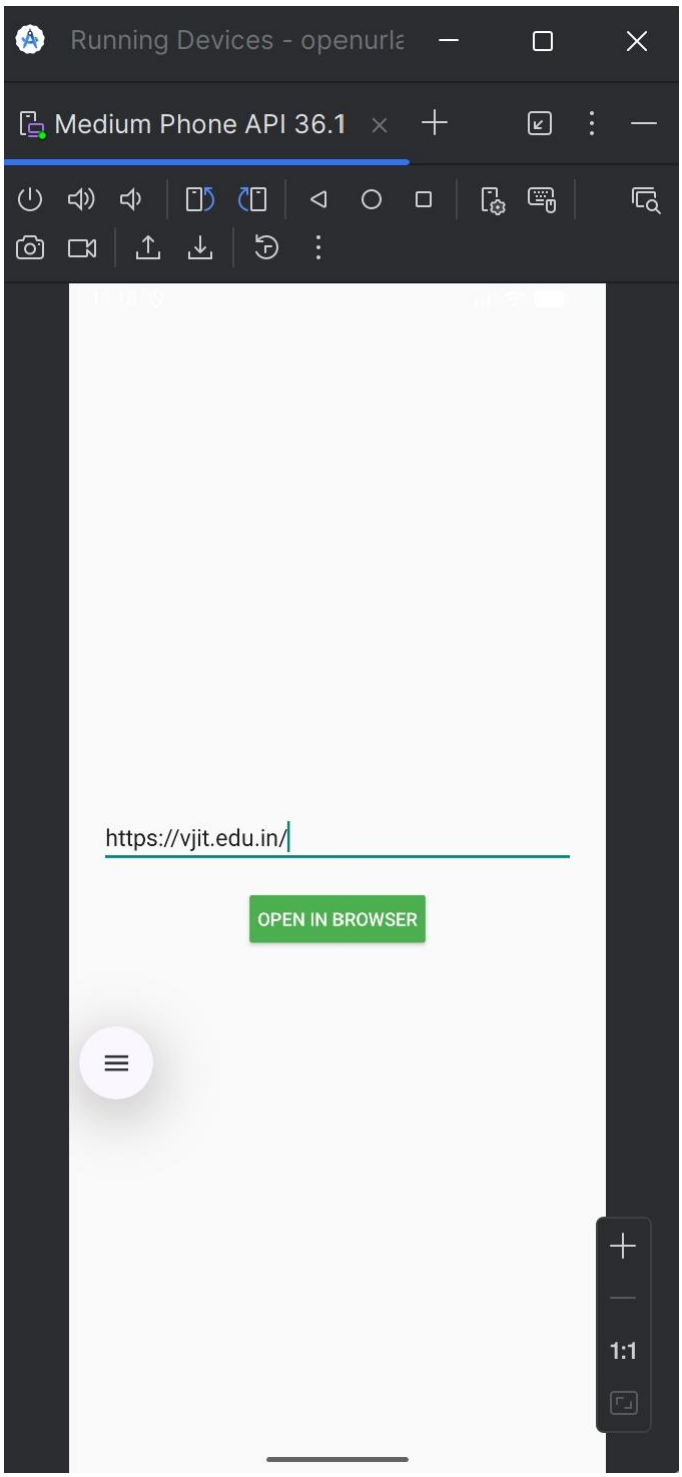
        if (u.isEmpty()) {
             Toast.makeText(MainActivity.this, "Please enter a
URL", Toast.LENGTH_SHORT).show();
             return;
}

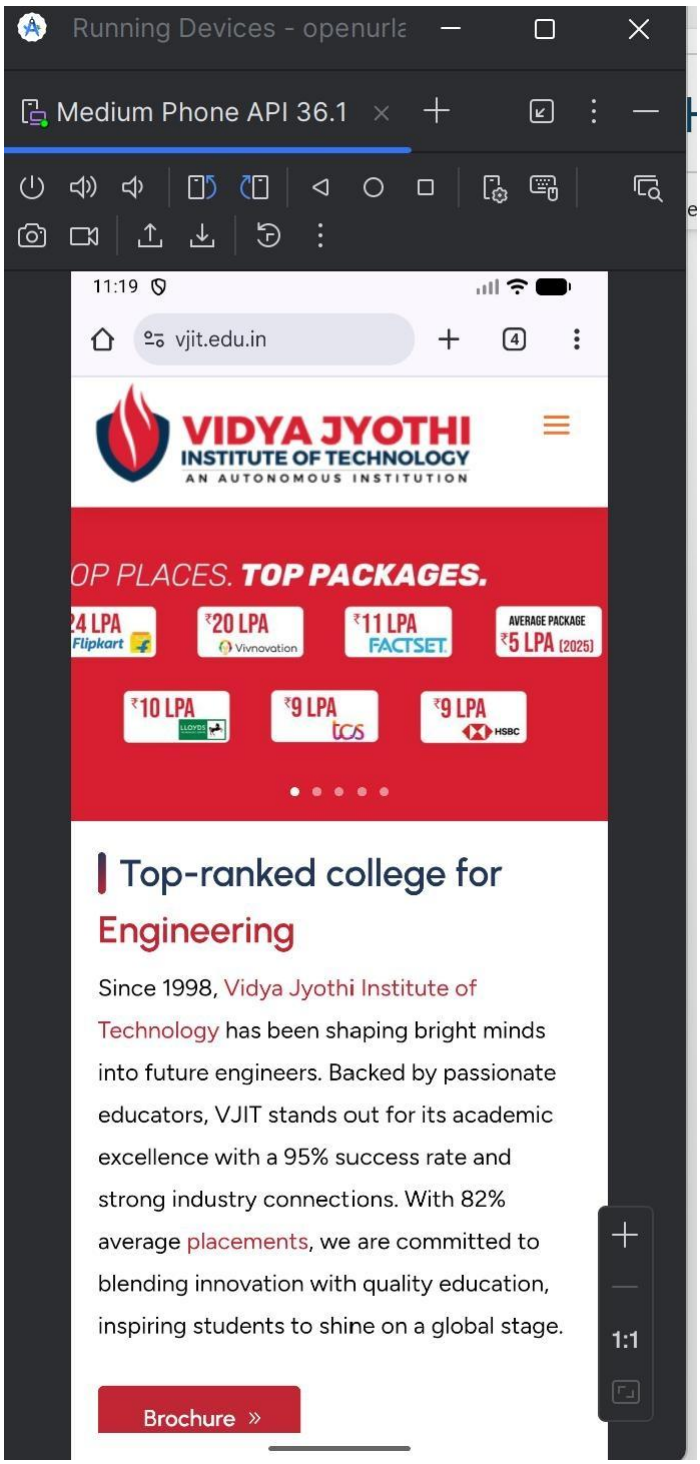
        // Add https:// if missing
        if (!u.startsWith("http://") && !u.startsWith("https://")) {
             u = "https://" + u; }
```

```
        try {
            Intent i = new Intent(Intent.ACTION_VIEW, Uri.parse(u));
startActivity(i);
        } catch (Exception e) {
            Toast.makeText(MainActivity.this, "Invalid
URL!", Toast.LENGTH_SHORT).show();
        }
}); }
}
```

Output:


Running Devices - openurla
Medium Phone API 36.1
Enter URL (e.g. https://www.google.com)
OPEN IN BROWSER
1:1


Running Devices - openurla
Medium Phone API 36.1
https://vjit.edu.in/
OPEN IN BROWSER
1:1


Running Devices - openurla
Medium Phone API 36.1
11:19
vjit.edu.in
VIDYA JYOTHI
INSTITUTE OF TECHNOLOGY
AN AUTONOMOUS INSTITUTION
OP PLACES. TOP PACKAGES.
24 LPA
Flipkart
₹20 LPA
Vivnovation
₹11 LPA
FACTSET
AVERAGE PACKAGE
₹5 LPA (2025)
₹10 LPA
₹9 LPA
tcs
₹9 LPA
HSBC
Top-ranked college for Engineering
Since 1998, Vidya Jyothi Institute of Technology has been shaping bright minds into future engineers. Backed by passionate educators, VJIT stands out for its academic excellence with a 95% success rate and strong industry connections. With 82% average placements, we are committed to blending innovation with quality education, inspiring students to shine on a global stage.
Brochure »
1:1

Week-11

Develop a native application that uses GPS location information.

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <Button
      android:id="@+id/b"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Get Location" />

   <TextView
      android :id="@+id/tv"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Location will appear here"
android:textSize="18sp"/> </LinearLayout>
```

MainActivity.java

```
package com.example.gpslocationapp;

//package com.example.gpslocationapp;

import android.Manifest;
import android.annotation.SuppressLint;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
```

```
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {
   private static final int PC = 100; private LocationManager lm;
   private TextView tv ;

   @Override
   protected void onCreate(Bundle bd) {
super.onCreate(bd);
      setContentView(R.layout.activity_main);

     Button b = findViewById(R.id.b); tv = findViewById(R.id.tv );

      b.setOnClickListener(v -> loc()); }

   private void loc () {
      if (ContextCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION) !=
PackageManager.PERMISSION_GRANTED) {
         ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.ACCESS_FINE_LOCATION},
PC);
      } else {
         lm = (LocationManager) getSystemService(LOCATION_SERVICE);
lm.requestLocationUpdates(LocationManager.GPS_PROVIDER, 0, 0, ll);
      }
   }

   private final LocationListener ll = new LocationListener() { @SuppressLint
("SetTextI18n")
      @Override
      public void onLocationChanged(Location loc1) {
         double lat = loc1.getLatitude();
         double lon = loc1.getLongitude();

         tv.setText("Latitude: " + lat + "\nLongitude: " + lon);
         lm.removeUpdates(this); // Optional: remove after first update
      }
      @Override public void onStatusChanged(String p, int s, Bundle ex) {} @Override
public void onProviderEnabled (@NonNull String p) {}
      @Override public void onProviderDisabled(@NonNull String p) {
```

```
        Toast.makeText(MainActivity.this, "Please enable
GPS", Toast.LENGTH_LONG).show();
      }
   };

   @Override
   public void onRequestPermissionsResult(int rc, @NonNull String[] per,
@NonNull int[] gr) {
      super.onRequestPermissionsResult(rc, per, gr); if (rc == PC) {
         if (gr.length > 0 && gr[0] ==
PackageManager.PERMISSION_GRANTED) {
            loc();
         } else {
            Toast.makeText(this, "Location permission
denied", Toast.LENGTH_SHORT).show();
         }
      }
   }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">

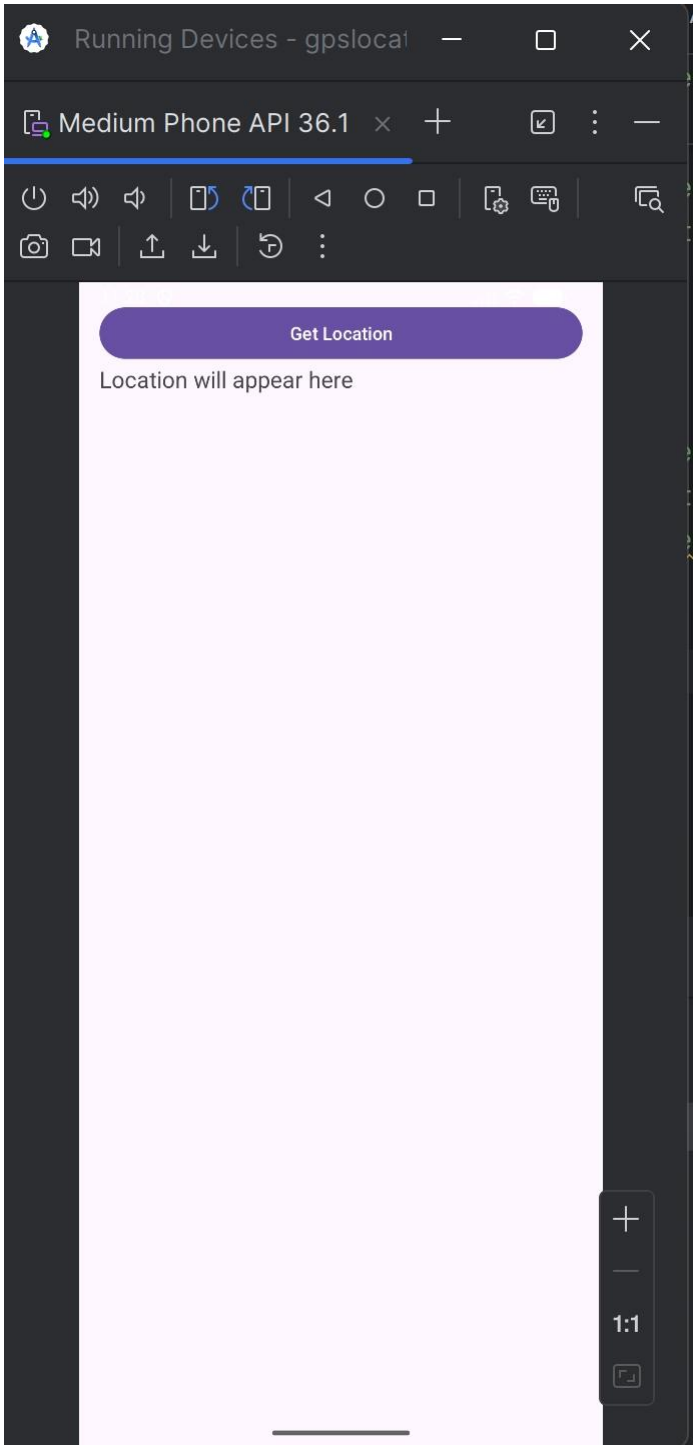
   <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"
/>
   <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"

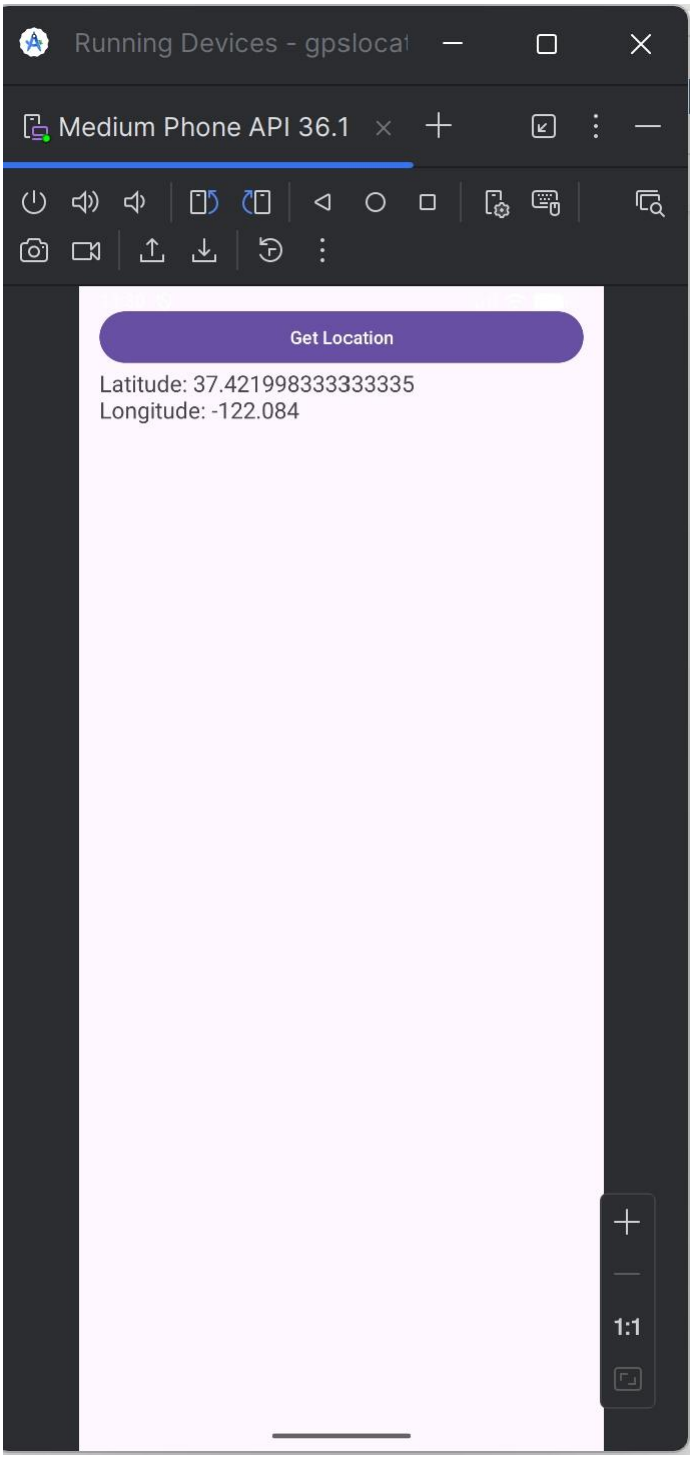
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.Gpslocationapp">
      <activity
         android :name =".MainActivity"
         android :exported ="true">
         <intent -filter>
            <action android:name="android.intent.action.MAIN" />
```

```
            <category android:name="android.intent.category.LAUNCHER" />
</intent -filter>
        </activity>
    </application>

</manifest>
```

Outputs:


Running Devices - gpsloca
Medium Phone API 36.1
Get Location
Location will appear here
1:1


Running Devices - gpsloca
Medium Phone API 36.1
Get Location
Latitude: 37.421998333333335
Longitude: -122.084
1:1